

Houghton's group H_3 has superpolynomial sofic profile

Seth Douglas

Nidhal Mghirbi

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Abstract

The sofic profile of a finite piece of a group measures the size of a permutation model that approximately preserves multiplication and separates distinct elements, at error $1/r$. We prove that Houghton's finitely presented elementary amenable group H_3 has a finite piece with profile at least $\exp(\Omega(r^c))$, where $c = 1/(\log_2 107 + 1)$, and at most $\exp(O(\sqrt{r} \log r))$. Thus a sofic group can have superpolynomial sofic profile, answering the last question of Cornulier's Problem 3.18. This also disproves his extension theorem as stated. We identify the missing dependence of the kernel chunk on the error scale and give a corrected bound. Every homomorphism from H_m , $m \geq 3$, to a group with polynomial sofic profile kills the finitary alternating subgroup; in particular this applies to birational groups. The proof combines certified polynomial fillings of far commutators in Johnson's presentation with an entropy count for approximately commuting copies of S_3 . Collection gives the full Dehn-function bound $\delta_{H_3}(n) = O(n^{\log_2 107+4})$. We also obtain quantitative unitary lower bounds and construct models that distribute the relator defect, improving the upper bound supplied by our permutation construction. The upper sofic bound holds for every fixed finite chunk of H_3 .

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1 Introduction

Sofic groups are the groups whose finite pieces can be modelled by permutations up to a small error, and hyperlinear groups those whose finite pieces can be modelled by unitary matrices. Quantitative versions ask, for groups that can be approximated, how large the models must be. Cornuier introduced the *sofic profile* $\sigma_E(r)$ of a finite piece (a chunk) E of a group: the least n for which E maps to the symmetric group Sym_n with multiplication defect $1/r$ and with distinct elements at normalized Hamming distance at least $1 - 1/r$ [2]. Amenable groups are sofic. More quantitatively, Cornuier derived from an extension theorem [2, Theorem 3.15] that every elementary amenable group has polynomial sofic profile [2, Example 3.16], and asked for methods to bound the sofic profile of explicit groups from below, whether some group has sofic profile that is unbounded and not linear in r , and whether such a group can be sofic [2, Problem 3.18].

Houghton's group H_3 is finitely presented and elementary amenable: it is an extension of the locally finite group $\text{FSym}(\mathbb{N})$ of finitary permutations by $\mathbb{Z}^2$ [5]. Section 2 defines it and a chunk E_H .

Theorem 1.1. *The chunk E_H of H_3 has sofic profile*

$$2^{r^{1/(\log 107+1)/1280-1/16}} \leq \sigma_{E_H}(r) \leq 2^{O(\sqrt{r} \log r)} \quad (r \geq 2).$$

The unweighted bound obtained from certified finite fillings already gives $\sigma_{E_H}(r) \geq 2^{r^{1/12}/270-1/16}$.

Here and below logarithms are to base two, and $\log 107 \approx 6.7415$. For $T \in M_D$ write $\tau_D(T) = D^{-1} \text{Tr } T$, $\|T\|_2^2 = \tau_D(T^*T)$ and $\|T\|_{HS}^2 = \text{Tr}(T^*T)$; the dimension subscript on τ is omitted when clear. For a density matrix ρ , $S(\rho) = -\text{Tr}(\rho \log \rho)$, and $h_2(p) = -p \log p - (1-p) \log(1-p)$, with $0 \log 0 = 0$.

Corollary 1.2. *Theorem 3.15 of [2] is false for the extension $1 \rightarrow \text{FSym}(\mathbb{N}) \rightarrow H_3 \rightarrow \mathbb{Z}^2 \rightarrow 1$ and the chunk E_H . So H_3 is an elementary amenable group whose sofic profile is not polynomial, contrary to [2, Example 3.16].*

Corollary 1.3 (Homomorphisms to polynomial-profile groups). *Let H_m denote the group of permutations of m rays that are eventually translations on each ray. Suppose every finite chunk of Γ has polynomial sofic profile. Every homomorphism $H_m \rightarrow \Gamma$, $m \geq 3$, kills the finitary alternating subgroup. Its image has commutator subgroup of order at most two, nilpotency class at most two, and an abelian subgroup of index at most $2^{\lfloor (m-1)/2 \rfloor}$; for H_3 this index is at most two.*

In particular these conclusions hold for $\Gamma = \text{Bir}_K(X)$, for any field K and absolutely irreducible finite-dimensional variety X over K . No group containing H_3 embeds into such a birational group.

The proof is in Section 2.1. In fact, every nontrivial normal subgroup of H_m contains the finitary alternating subgroup (Lemma 2.2). Thus every homomorphic image is either a faithful copy of H_m , with superpolynomial sofic profile, or a residually finite, virtually abelian quotient with bounded profile on each finite chunk (Proposition 2.3).

We locate the error in Section 7: the extension theorem asserts a finite piece of the kernel that does not depend on the error parameter, but the construction it relies on needs kernel elements conjugated along a Følner set whose size grows with the inverse error. Tracking that dependence gives a correct, scale-dependent extension bound (Proposition 7.2). Since H_3 is amenable, and hence sofic, Theorem 1.1 also answers the last question of Cornuier's Problem 3.18: a sofic group can have sofic profile that is unbounded and not linear in r .

Slofstra constructed a finitely presented group with superpolynomial hyperlinear and sofic profile [7], answering the first question; that group is not amenable, and its soficity and even hyperlinearity were left unresolved there. To our knowledge H_3 is the first amenable group shown to have superpolynomial sofic profile.

The lower bound holds for unitary approximations as well. Write $P = \langle a, b, \alpha \mid r_1, \dots, r_5 \rangle$ for Johnson’s presentation (Section 2) and call unitaries $\phi(a), \phi(b), \phi(\alpha) \in U(D)$ an *e-representation* if every relator satisfies $\|\phi(r_i) - I\|_2 \leq e$ in the normalized Hilbert–Schmidt norm.

Theorem 1.4. *Let $\beta_H = 1/(2 \log 107 + 1) \approx 1/14.48$, $K' = 750 \cdot 135 \cdot 3^{\log 107}$ and $e > 0$. Every e -representation of P on $\mathbb{C}^D$ in which the three-cycle $\gamma = \alpha a \alpha a^{-1}$ satisfies $\|\phi(\gamma) - I\|_2 \geq 1/2$ has $\log D \geq (K'e)^{-2\beta_H}/16 - 1/16$.*

Conversely, spreading the single defect of our permutation models over many cells gives such representations with $\log D = O(e^{-2/3})$, while our permutation models give only $O(e^{-1})$ (Corollary 6.7).

Corollary 1.5 (Exact finite-dimensional representations). *Every homomorphism $H_m \rightarrow U(D)$, $m \geq 3$, kills the finitary alternating subgroup. Every irreducible finite-dimensional unitary representation of H_m has dimension either 1 or $2^{\lfloor (m-1)/2 \rfloor}$, and both dimensions occur.*

The proof is in Section 2.1.

How the proofs go.

- *Far commutators have polynomial area* (Section 3). The commutator of two conjugates of the basic transposition at distance R has area at most $135R^{\log 107}$ in Johnson’s presentation, by a doubling self-similarity of H_3 whose finite relations are certified by machine. Buffered collection upgrades these estimates to a polynomial bound on the full Dehn function (Theorem 3.3), improving the exponential upper bound of [4].
- *Displaced copies of S_3 need dimension* (Section 4). An approximate representation gives, through the far commutators, many approximately commuting copies of S_3 whose three-cycles are displaced from the identity. Rounding each copy to an exact one and counting the entropy needed to make a state invariant forces dimension exponential in their number.
- *Transport* (Section 5). For permutations, relator errors add up in measure over a van Kampen diagram; for unitaries, in norm. This converts the area bound into the two exponents.
- *Upper bounds* (Section 6). Permutation models on a torus of clock positions that fail at a single corner, and unitary models that spread that failure over many cells by small rotations.

Related work. Quantitative metric approximations of groups are surveyed in [9]; the hyperlinear profile and its relation to nonlocal games are studied in [8, 7], and hyperlinear approximations of amenable groups come from sofic ones [10]. Our method follows Slofstra’s in combining stability of approximate representations of finite groups, here an entropy bound in place of the stability theorems of Gowers–Hatami [11] and De Chiffre–Ozawa–Thom [12], with an isoperimetric input. Houghton’s groups have an exponential isoperimetric inequality [4] and are not co-Hopfian [6]; we use the doubling endomorphism behind the latter. The extension theorem is attributed in [2] to the construction of Elek and Szabó [3]. A companion

paper [14] applies the lower bounds to the memory needed for the repeated use of a quantum channel built from H_3 .

Glebe [15, Corollary 4.5] proves that H_m , $m \geq 3$, is not stable in the unnormalized Schatten p -norms, $1 < p \leq \infty$, by a cup-product obstruction. That qualitative nonstability result differs from the normalized-error dimension profiles considered here: we require separation of a fixed finite piece and bound the dimension of every approximate model. In particular, changing the matrix-norm normalization is not innocuous in a comparison of these statements.

Organization. Section 2 defines H_3 , the chunk and the profiles. Section 3 proves the area bounds, Section 4 the dimension bounds for approximate representations, and Section 5 the lower bounds of Theorems 1.1 and 1.4. Section 6 gives the upper bounds, Section 7 the correction, and Section 8 open problems. Appendix A gives the fillings.

2 Houghton's group, chunks and profiles

Let $Y = \{(i, j) : i \in \{1, 2, 3\}, j \geq 1\}$ be three rays. Permutations act on the right and gh means g followed by h . Let a send $(1, 1) \mapsto (2, 1)$, $(1, j) \mapsto (1, j - 1)$ for $j \geq 2$ and $(2, j) \mapsto (2, j + 1)$, fixing ray 3; let b be defined in the same way with rays 2 and 3 exchanged; let α be the transposition of $(1, 1)$ and $(1, 2)$. These generate H_3 , the group of permutations of Y that are eventually translations on each ray. They satisfy

$$r_1 = \alpha^2, \quad r_2 = (\alpha a \alpha a^{-1})^3, \quad r_3 = [\alpha, a^2 \alpha a^{-2}], \quad r_4 = aba^{-1}b^{-1}\alpha^{-1}, \quad r_5 = a\alpha a^{-1}b\alpha^{-1}b^{-1}, \quad (1)$$

with $[x, y] = xyx^{-1}y^{-1}$; each identity can be checked directly on the finitely many points where the word moves anything. By a theorem of Johnson (see [4, Theorem 2.8]) these relators present H_3 . The lower bound below uses only that $r_1, \dots, r_5$ hold in H_3 , the far-commutator areas of Section 3, and that the three-cycle $\gamma = \alpha(a\alpha a^{-1})$ on $(1, 1), (1, 2), (1, 3)$ is nontrivial. The short exact sequence $1 \rightarrow \text{FSym}(\mathbb{N}) \rightarrow H_3 \rightarrow \mathbb{Z}^2 \rightarrow 1$ shows that H_3 is elementary amenable.

We recall Cornuier's definitions [2, Definitions 3.2 and 3.4]. A *chunk* of a group G is a finite subset $E \ni 1$ with the partial multiplication $xy = z$ whenever $x, y, z \in E$ and $xy = z$ in G . For a map $f : E \rightarrow \text{Sym}_n$ with $f(1) = \text{id}$, let d_n be the normalized Hamming distance on Sym_n . The map is an ε -*morphism* if $d_n(f(xy), f(x)f(y)) \leq \varepsilon$ whenever xy is defined in E , and it is $(1 - \varepsilon)$ -*expansive* if $d_n(f(x), f(y)) \geq 1 - \varepsilon$ for distinct $x, y \in E$. The sofic profile is

$$\sigma_E(r) = \min\{n : \text{there is a } (1 - r^{-1})\text{-expansive } r^{-1}\text{-morphism } E \rightarrow \text{Sym}_n\}, \quad r > 1.$$

A group has polynomial sofic profile if every chunk has $\sigma_E \preceq$ some polynomial [2, Definition 3.10].

Let $E_H \subset H_3$ be the symmetric chunk consisting of $1, a^{\pm 1}, b^{\pm 1}, \alpha^{\pm 1}$, all prefixes of the relators $r_1, \dots, r_5$ and of the word $\alpha a \alpha a^{-1}$ for γ , and the inverses of these elements.

For permutations, relations can be transported in measure: the normalized Hamming distance d_n is bi-invariant, so a null word that is a product of A conjugates of relators has $d_n(\phi(w), \text{id}) \leq A\delta$ when every relator has $d_n(\phi(r_i), \text{id}) \leq \delta$. For unitaries only $\|\phi(w) - I\|_2 \leq A \max_i \|\phi(r_i) - I\|_2$ is available in general, since $\|(e^{i\theta} I)^A - I\|_2$ grows like $A\theta$.

2.1 Images and finite-dimensional representations

Proof of Corollary 1.3. Let $f : H_3 \rightarrow \Gamma$ and suppose $f(\gamma) \neq 1$, where $\gamma = \alpha a \alpha a^{-1}$ is the three-cycle used in the lower bound. The image chunk $f(E_H)$ retains every multiplication

relation among the relator prefixes, even if other labels become identified. An expansive r^{-1} -morphism of this chunk to Sym_D therefore gives generator permutations with relator errors at most $23/r$ and displacement of γ at least $1 - 5/r$. For $r \geq 10$, the proof of Theorem 1.1, using Proposition 5.1, applies unchanged: it only needs these relations and the displacement of this one three-cycle. Thus $\log D \geq r^{1/(\log 107+1)}/1280 - 1/16$, contradicting the polynomial profile of $f(E_H)$. Hence f kills γ , and then all finitary three-cycles by conjugation. These generate the finitary alternating subgroup.

For H_m , restrict to the copy of H_3 on the first three rays and conjugate by finitary permutations of all m rays. The same conclusion follows. The quotient by the finitary alternating subgroup is a central extension of $\mathbb{Z}^{m-1}$ by C_2 : its kernel is the parity of a finitary permutation, which conjugation preserves. Its commutators are therefore central of order at most two. Put $d = m - 1$. Commutation defines an alternating bilinear form on $\mathbb{F}_2^d$ with values in C_2 : it is independent of the lifts, and $[x^2, y] = [x, y]^2 = 1$. Such a form has an isotropic subspace of dimension at least $\lceil d/2 \rceil$. Indeed, if the form is nonzero, split off a nondegenerate plane spanned by e, f with pairing one, apply induction to its orthogonal complement, and adjoin e ; if it vanishes, take the whole space. The inverse image of this isotropic subspace in the central extension is abelian and has index at most $2^{\lceil d/2 \rceil}$. Its image under any quotient map has no larger index. The other stated properties also pass to every image.

Finally, the independent Corollary 4.5 of [2] gives polynomial profile for every chunk of $\text{Bir}_K(X)$; replace X by a nonempty affine open if necessary. This proves the birational assertions. $\square$

Proof of Corollary 1.5. For H_3 , let Δ be the displacement of γ under an exact representation. Restrict to N disjoint copies of S_3 as in Section 4. Their three-cycles are conjugate and have displacement Δ , and different copies commute. The exact case of Lemma 4.2 gives $\log D \geq N\Delta^2/3$ for every N , so $\Delta = 0$. Conjugation then kills every finitary three-cycle; restriction to an embedded H_3 gives the same conclusion for H_m . The central-extension argument in the proof of Corollary 1.3 gives an abelian subgroup of the image of index $q \leq 2^{\lfloor (m-1)/2 \rfloor}$. Choose a common eigenvector of its commuting unitary operators. Its orbit lies in the span of its images under q coset representatives, an invariant space of dimension at most q . Irreducibility makes this the whole representation space. The exact dimensions and their attainment follow from Proposition 2.1. $\square$

Proposition 2.1 (Parity quotient and sharp image bounds). *Put $d = m - 1$, $r = \lfloor d/2 \rfloor$, and let A_{fin} be the finitary alternating subgroup. The parity quotient has presentation*

$$H_m/A_{\text{fin}} = \langle t_1, \dots, t_d, z \mid z^2 = 1, \ z \text{ central}, \ [t_i, t_j] = z \ (i < j) \rangle.$$

The abelian-index bound in Corollary 1.3 is attained by a finite image. The irreducible unitary representations have dimensions 1 and 2^r , and both occur.

Proof. Indeed, take t_i to translate from ray 1 to ray $i + 1$. Directly in the ray action, every $[t_i, t_j]$ is the same basic transposition α . These translations generate H_m : they generate the translation lattice, and conjugates of α give adjacent transpositions along the connected union of rays, hence every finitary permutation. Their classes and $z = \alpha A_{\text{fin}}$ satisfy the displayed relations. Every abstract word collects to $z^\epsilon t_1^{n_1} \cdots t_d^{n_d}$. If its image is trivial, translation forces all $n_i = 0$, and parity forces $\epsilon = 0$. Thus there are no further relations.

By the preceding factorization argument and dimension bound, it remains to consider an irreducible representation of this quotient. Schur's lemma gives $z = \pm I$. For $z = I$ the generators commute, so the dimension is one. For $z = -I$, their matrices T_i pairwise anticommute, and T_i^2 is central. Rescale each T_i by a scalar phase so that $T_i^2 = I$. The

operators $H_j = iT_{2j-1}T_{2j}$, $1 \leq j \leq r$, are commuting Hermitian involutions. From a common eigenvector, products of the T_{2j-1} reach all 2^r joint eigenvalue patterns, since each flips exactly the j th sign. These vectors are nonzero and orthogonal. The dimension is at least 2^r , hence exactly 2^r by the preceding upper bound.

For attainment, use $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ and $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. On $(\mathbb{C}^2)^{\otimes r}$ take

$$\Gamma_{2j-1} = Z^{\otimes(j-1)} \otimes X \otimes I^{\otimes(r-j)}, \quad \Gamma_{2j} = Z^{\otimes(j-1)} \otimes Y \otimes I^{\otimes(r-j)}.$$

If $d = 2r + 1$, append $\Gamma_{2r+1} = Z^{\otimes r}$. These pairwise anticommute and square to I , so $t_i \mapsto \Gamma_i$, $z \mapsto -I$ defines a representation. The first pair generates all matrices on the first qubit; removing its Z -strings from the next pair and iterating proves irreducibility. Its image F lies in the finite Pauli group. Any abelian subgroup $B \leq F$ has a common eigenvector, whose coset orbit and irreducibility give $[F : B] \geq 2^r$. Corollary 1.3 gives the reverse bound, because a finite group has bounded chunk profiles. Thus its abelian-index bound is sharp already for a finite image. $\square$

Lemma 2.2 (Normal subgroups). *For $m \geq 3$, every nontrivial normal subgroup of H_m contains A_{fin} .*

Proof. First, A_{fin} is simple. If $N \triangleleft A_{\text{fin}}$ is nontrivial, choose $1 \neq x \in N$, a moved point p , and distinct points r, s outside the support of x . Put $q = px^{-1} \neq p$ and $c = (p \ r \ s)$. With our right-action convention,

$$[x, c] = (q \ r \ s)(p \ s \ r) = (q \ p \ s).$$

Thus N contains a three-cycle. All three-cycles are conjugate in A_{fin} : any finitary conjugator can be made even by composing with a transposition on two unused points. Since three-cycles generate A_{fin} , we have $N = A_{\text{fin}}$.

Now let $K \triangleleft H_m$. Both K and A_{fin} are normal, so $[K, A_{\text{fin}}] \subseteq K \cap A_{\text{fin}}$. If the intersection is nontrivial, simplicity gives the conclusion. Otherwise K centralizes A_{fin} . Its centralizer in the full symmetric group on the rays is trivial: if a permutation x moves p , set $q = px^{-1} \neq p$ and choose distinct $r, s \notin \{p, q\}$. The conjugate $x(p \ r \ s)x^{-1}$ has q in its support, whereas $(p \ r \ s)$ does not, so x cannot centralize that three-cycle. Hence $K = 1$ in this case. $\square$

Proposition 2.3 (Classification of polynomial-profile images). *Let $f : H_m \rightarrow \Gamma$, $m \geq 3$, and put $F = f(H_m)$. The following are equivalent: every finite chunk of F has polynomial sofic profile; every finite chunk of F has bounded sofic profile; F is residually finite; $f(A_{\text{fin}}) = 1$; and f is not injective. Every such image is virtually abelian, with the sharp index bound of Proposition 2.1. Moreover, A_{fin} is the intersection of the kernels of all finite quotients of H_m , and also of all finite-dimensional unitary representations.*

Proof. Lemma 2.2 gives the equivalence of noninjectivity and $f(A_{\text{fin}}) = 1$. The polynomial-profile condition implies $f(A_{\text{fin}}) = 1$ by Corollary 1.3. If f kills A_{fin} , write its image using the generators of Proposition 2.1, and let $C = \langle f(z), f(t_1)^2, \dots, f(t_{m-1})^2 \rangle$. This is a finitely generated central abelian subgroup of finite index in F . An element outside C is detected by F/C . A nonidentity element $g \in C$ is outside kC for some positive integer k , by the structure theorem for finitely generated abelian groups. The subgroup kC is central and F/kC is finite, so it detects g . Thus F is residually finite. For a fixed finite chunk, residual finiteness separates its finitely many pairwise differences in a finite product of quotients. The regular action of that quotient is an exact, fully separating permutation model at every accuracy, hence the profile is bounded. Bounded implies polynomial. Finally every finite quotient kills A_{fin} , while the parity quotient is residually finite by the argument just given. Finite regular representations separate the same quotient; Corollary 1.5 supplies the reverse inclusion for unitary kernels. $\square$

3 Polynomial area of far commutators

Let $P = \langle a, b, \alpha \mid r_1, \dots, r_5 \rangle$, put $\sigma_j = a^j \alpha a^{-j}$, and let $A^*(M)$ be the largest area in P of a commutator $[\sigma_p^{\pm 1}, \sigma_q^{\pm 1}]$ with $0 \leq p, q \leq M$ and $|p - q| \geq 2$. These commutators are trivial in H_3 because the transpositions have disjoint supports.

Theorem 3.1 (Polynomial far-commutator area). $A^*(M) \leq M^{11}$ for every $M \geq 2$.

Outline; the full proof is in Appendix A. Let ψ be the endomorphism of the free group on a, b, α with $\psi(a) = a^2$, $\psi(b) = b^2$ and $\psi(\alpha) = [a^2, b^2]$. On H_3 it is induced by the diagonal action on two copies of Y , identified with Y by the bijection $(1, j, e) \mapsto (1, 2j - 1 + e)$ and $(i, j, e) \mapsto (i, 2j - e)$ for $i = 2, 3$ and $e \in \{0, 1\}$. The resulting group endomorphism, not this bijection of sets, is injective and not surjective (compare [6]). The proof has four steps.

- (1) *ψ lifts to P with bounded area multiplier.* Each $\psi(r_i)$ is a null word in P , of area at most 0, 17, 182, 864, 359 for $i = 4, 1, 5, 3, 2$. Hence $\text{Area}_P(\psi(w)) \leq 864 \text{Area}_P(w)$.
- (2) *Two cheap letters.* With $t = b^{-1}a$ and $\rho = a^{-1}b^{-1}ab$, the letter α commutes with t at cost 1 and with ρ at cost at most 5, and $\{t, \rho\}$ is invariant under ψ up to one relation, $b^{-1}\rho b = t\rho t^{-1} \cdot \rho \cdot t\rho t^{-1}$, of area at most 14. The identities $\psi(t) = t\rho t$, $t\rho = b^{-1}tb$ and $\psi(\rho) = t^{-1}\rho^b \rho^{b^2} t \rho \rho^b$ hold in the free group.
- (3) *A doubling recursion.* The words $X_n = b^{-n}a^n$ satisfy $X_{2n} = \psi(X_n)$ and $X_{2n+1} = b^{-1}X_{2n}bt$ freely, and in H_3 act as a reversal of $[1, n]$ followed by translation on the line formed by rays 2 and 3. Writing X_n as a word W_n in t, ρ costs $R(n)$ cells, with $R(2n) \leq 864 R(n) + 84 \#_\rho(W_n)$ and a comparable odd step. Unrolling gives $R(n) \leq 1.35 n^{\log_2 864}$.
- (4) *From X_n to far commutators.* $[\alpha, X_n]$ expands into one conjugate of $[\alpha, t]^{\pm 1}$ or $[\alpha, \rho]^{\pm 1}$ per letter of W_n , so its area is at most $2R(n) + 5|W_n|$; a far commutator $[\sigma_0, \sigma_m]$ reduces to it at linear extra cost. This gives $A^*(m) \leq 5.4 m^{\log_2 864} + 10 m^{\log_2 323} + 18m \leq m^{11}$ for $m \geq 5$, and $A^*(2), A^*(3), A^*(4) \leq 1, 21, 47$.

□

Certificates. Every finite relation used in the proof is supplied in machine-checkable form: as an explicit list of cells (z_k, r_{i_k}, s_k) such that the relation equals $\prod_k z_k r_{i_k}^{s_k} z_k^{-1}$ in the free group on a, b, α , which a short independent program verifies by free reduction after recomputing the relation from its definition. The certified relations and their cell counts are: $A(2) = 1$, $A(3) \leq 21$, $A(4) \leq 47$, $A(5) \leq 79$, $A(6) \leq 157$; $E(0..3) \leq 1, 4, 27, 76$; $U(1..3) \leq 1, 5, 32$; $B(2), B(3) \leq 9, 12$; the relation of step (2) with 14 cells; $[\alpha, \rho]$ with 5; $\psi(\alpha) = \sigma_1 \sigma_2 \sigma_0 \sigma_1$ with 6; the images $\psi(r_1), \psi(r_2), \psi(r_3), \psi(r_5)$ with 17, 359, 864, 182; and the rewriting of $\psi(\rho)$ as a word in t and ρ , which uses the relation of step (2) six times, with 84. The recursion of step (3) has also been executed mechanically, transporting certificates through ψ cell by cell: for $n \leq 9$ the certificates of $X_n = W_n$ have 0, 0, 14, 84, 392, 995, 2493, 5298, 11878 cells, each at most the recursion bound, and the resulting far commutators satisfy $A(m) \leq 73\,121$ for $m \leq 11$, far below m^{11} . Appendix A gives the fillings in words; the ancillary files contain the generator, the verifier and the exported certificates. The exported file holds every finite relation listed above and the recursion for $n \leq 5$, which the independent verifier checks; the instances $6 \leq n \leq 9$ were executed by the generator but are not in the file, and the generator exports them for the verifier on request. The proof uses only the finite relations and the hand induction of Appendix A.

Keeping a derived nine-cell relation atomic during substitution gives a sharper bound.

Theorem 3.2 (Reweighted far-commutator area). $A^*(M) \leq 135 M^{\log 107}$ for every $M \geq 2$.

The proof is in Appendix A. The same area estimates also control arbitrary null words.

Theorem 3.3 (Polynomial Dehn function). *Let $\delta_P(n)$ be the maximum area of a null word of length at most n in Johnson's presentation. There is an absolute constant C_0 such that*

$$\delta_P(n) \leq C_0 n^4 (1 + A^*(4n + 2)) = O(n^{\log 107 + 4}) \quad (n \geq 1).$$

The unweighted estimate of Theorem 3.1 alone gives $\delta_P(n) = O(n^{15})$.

The proof in Appendix B collects a buffered null word into adjacent transpositions, then fills it using a uniform Coxeter insertion algorithm. This full Dehn bound does not improve the profile exponents: the direct far-commutator estimate is sharper for the words used there. The explicit integer weights give exponent $\log 107 \approx 6.7415$. The auxiliary substitution matrix has spectral radius $\rho \approx 106.3385$ and also gives the asymptotic bound $O(M^{\log \rho})$ (Appendix A). Neither is an optimal-area claim. The area is at least quadratic.

Proposition 3.4 (Quadratic lower bound). *For odd $k \geq 9$, $\text{Area}(C_k) \geq (k - 7)(k - 5)/8$, where $C_k = [\alpha, a^k \alpha a^{-k}]$. Hence $A^*(R) \geq R^2/16$ for every integer $R \geq 23$.*

Proof. In a permutation model in which every relator moves at most a fraction δ of the points, a null word of area A moves at most a fraction $A\delta$ (transport in measure, Section 2). In the corner models of Proposition 6.1, $\delta \leq M^{-2}$ while C_{2M-1} moves a fraction at least $(1 - s^{-2})(M - 4)(M - 3)/(2M^2)$. Hence $\text{Area}(C_{2M-1}) \geq (1 - s^{-2})(M - 4)(M - 3)/2$ for every s , which gives the claim with $k = 2M - 1$. For integer $R \geq 23$ apply it to the largest odd $k \leq R$. $\square$

Thus the present uniform-area transport and sector estimate cannot give more than the exponent $1/3$ in r for permutation approximations, and $2/5$ in $1/e$ for unitary approximations with relator defect e (Section 8). This is a limit of that argument, not of every possible use of far commutators.

4 Stability and counting

The full S_3^N sector bound below feeds the permutation-profile estimate; the phase-robust version later in the section feeds the unitary estimate and the companion's weighted extraction. The entropy arguments use monotonicity of the Holevo quantity under partial trace, a consequence of strong subadditivity [1].

4.1 An entropic sector bound

Lemma 4.1 (Projection products). *Let $P_1, \dots, P_m$ be orthogonal projections on a Hilbert space and $\|x\| = 1$. Then $1 - \text{Re}\langle x, P_m \cdots P_1 x \rangle \leq 2 \sum_j \|(I - P_j)x\|^2$.*

Proof. Put $x_0 = x$, $x_j = P_j x_{j-1}$ and $r_j = x_{j-1} - x_j \in \text{ran}(I - P_j)$. Then $\sum_j \|r_j\|^2 = 1 - \|x_m\|^2 =: b$, and $a = 1 - \text{Re}\langle x, x_m \rangle = \text{Re} \sum_j \langle (I - P_j)x, r_j \rangle \leq \sqrt{Eb}$ with $E = \sum_j \|(I - P_j)x\|^2$. Since $0 \leq \|x - x_m\|^2 = 2a - b$, we get $a \leq \sqrt{2Ea}$, that is $a \leq 2E$. $\square$

Lemma 4.2 (Entropic sector bound). *Let $u_1, \dots, u_N : S_3 \rightarrow U(D)$ be unitary representations, not assumed to commute, such that for a three-cycle c*

$$\|u_i(c) - I\|_2 \geq \Delta \quad (1 \leq i \leq N), \quad \|u_i(g)u_j(h) - u_j(h)u_i(g)\|_2 \leq \eta \quad (i \neq j, \ g, h \in S_3).$$

Put $t = \sqrt{2N} \eta$. If $t \leq \Delta$ and $t^2 \leq 5/6$, then

$$\log D \geq N \left[\frac{(\Delta - t)^2}{3} - h_2(t^2) - t^2 \log 5 \right].$$

Proof. Let $G = S_3^N$ and $f(g) = u_1(g_1) \cdots u_N(g_N)$. For h in factor i , $\|f(gh) - f(g)u_i(h)\|_2 = \|S^*XS - X\|_2$ with $X = u_i(h)$ and $S = u_{i+1}(g_{i+1}) \cdots u_N(g_N)$, and $\|S^*XS - X\|_2^2 = 2 - 2\operatorname{Re} \tau(X^*S^*XS)$. The averages $T_j(Y) = \frac{1}{6} \sum_a u_j(a)^* Y u_j(a)$ are orthogonal projections on M_D with the normalized Hilbert–Schmidt inner product, the average of S^*XS over g is $T_N \cdots T_{i+1}(X)$, and $\|(I - T_j)X\|_2^2 = \frac{1}{12} \sum_a \|X, u_j(a)\|_2^2 \leq \eta^2/2$. Lemma 4.1 therefore gives

$$\mathbb{E}_g \|f(gh) - f(g)u_i(h)\|_2^2 \leq 2(N - i)\eta^2 \leq t^2.$$

Let R be the right regular representation of G and $W : \mathbb{C}^D \rightarrow \ell^2(G) \otimes \mathbb{C}^D$ the isometry $Wv = |G|^{-1/2} \sum_g |g\rangle \otimes f(g)v$. Then $\|R(h)W - Wu_i(h)\|_{HS} \leq \sqrt{D}t$ for h in factor i . Put $P = WW^*$ and $\rho = P/D$, a state of entropy $\log D$. For the three-cycle c_i of factor i and $Z_i = 2I - R(c_i) - R(c_i)^*$,

$$\operatorname{Tr}(\rho Z_i) = D^{-1} \|(R(c_i) - I)W\|_{HS}^2 \geq (\Delta - t)^2.$$

Let $\mathcal{E}_i(X) = \frac{1}{6} \sum_h R(h_i)XR(h_i)^*$, with h_i the element h placed in factor i ; these conditional expectations commute. Since $(I - P)W = 0$, $\operatorname{Tr}[(I - P)\mathcal{E}_i(\rho)] \leq t^2$. The state $\mathcal{E}_i(\rho)$ is an average of six states of rank D , one of them ρ , so after pinching by P and $I - P$ its two blocks have rank at most D and $5D$. Pinching does not decrease entropy and $x \mapsto h_2(x) + x \log 5$ increases on $[0, 5/6]$, so $S(\mathcal{E}_i(\rho)) \leq \log D + h_2(t^2) + t^2 \log 5$. For a conditional expectation $S(\mathcal{E}(\sigma)) - S(\sigma) = D(\sigma \|\mathcal{E}(\sigma))$; since the $\mathcal{E}_j$ commute, monotonicity of relative entropy under $\mathcal{E}_{i-1} \cdots \mathcal{E}_1$ bounds the entropy increment of $\mathcal{E}_i$ applied after $\mathcal{E}_{i-1} \cdots \mathcal{E}_1$ by $D(\rho \|\mathcal{E}_i \rho)$. Hence $\bar{\rho} = \mathcal{E}_N \cdots \mathcal{E}_1(\rho)$ satisfies $S(\bar{\rho}) \leq \log D + N[h_2(t^2) + t^2 \log 5]$.

The state $\bar{\rho}$ commutes with $R(G)$, so $\bar{\rho} = \bigoplus_\lambda p_\lambda (I_{d_\lambda}/d_\lambda \otimes \sigma_\lambda)$ over the irreducible representations λ of S_3^N , and $S(\bar{\rho}) \geq \sum_\lambda p_\lambda \log d_\lambda = \sum_\lambda p_\lambda k_\lambda$, where k_λ is the number of factors of λ carrying the two-dimensional representation of S_3 . The operator Z_i is central in the group algebra, so $\operatorname{Tr}(\bar{\rho} Z_i) = \operatorname{Tr}(\rho Z_i)$, and it acts as $3I$ on the sectors whose factor i is two-dimensional and as 0 on the others. Hence $N(\Delta - t)^2 \leq \sum_i \operatorname{Tr}(\bar{\rho} Z_i) = 3 \sum_\lambda p_\lambda k_\lambda \leq 3S(\bar{\rho})$. $\square$

Lemma 4.3 (Exact S_3 rounding). *If $A, B \in U(D)$ satisfy $\|A^2 - I\|_2, \|B^2 - I\|_2, \|(AB)^3 - I\|_2 \leq e$, there is a unitary representation u of $S_3 = \langle s, t \mid s^2, t^2, (st)^3 \rangle$ on $\mathbb{C}^D$ with $\|u(s) - A\|_2 \leq e$, $\|u(t) - B\|_2 \leq 8e$ and $\|u(st) - AB\|_2 \leq 9e$, and each normal form $1, s, t, st, ts, sts$ evaluated in u is within $10e$ of the same word in A, B .*

Proof. Round the spectra of A and B to the nearer of ± 1 . For $|z| = 1$ this moves z by at most $|z^2 - 1|$, so the resulting Hermitian unitaries S, T satisfy $\|S - A\|_2, \|T - B\|_2 \leq e$. Then $C = ST$ satisfies $SCS = C^*$ and $\|C^3 - I\|_2 \leq 7e$. Round the spectrum of C to the nearest cube root of unity, choosing 1 at ties except for the following explicit exception: send -1 to 1, although it is not the nearest root there. At this exceptional point the displacement is $2 = |(-1)^3 - 1|$. This rule commutes with complex conjugation and moves z by at most $|z^3 - 1|$, so C_0 satisfies $C_0^3 = I$, $SC_0S = C_0^*$ and $\|C_0 - C\|_2 \leq 7e$. Then $u(s) = S$ and $u(t) = SC_0$ are involutions with $u(s)u(t) = C_0$, and the bounds follow from the triangle inequality. $\square$

4.2 A phase-robust form

The count of Lemma 4.2 can be run on products of three-cycles alone. It then needs no relation between the transpositions of different copies, so a coherent phase in their commutators does no harm. Write $\kappa(u, v) = \|uv - vu\|_2$ and $G(t) = h_2(\min\{\frac{2}{3}t, \frac{2}{3}\}) + \min\{\frac{2}{3}t, \frac{2}{3}\}$.

Theorem 4.4 (Phase-robust sector bound). *Let $c_i, x_i \in U(D)$, $1 \leq i \leq N$, satisfy $c_i^3 = x_i^2 = I$ and $x_i c_i x_i = c_i^{-1}$, with no condition on the commutators of x_i and x_j . Put $\Delta_i = \|c_i - I\|_2$ and*

$f(g) = c_1^{g_1} \cdots c_N^{g_N}$ for $g \in \mathbb{Z}_3^N$, let $g^{(i)}$ be g with its i -th coordinate negated, and, for uniform g ,

$$\hat{a}_i^2 = \mathbb{E}_g \|f(g + e_i) - f(g)c_i\|_2^2, \quad \hat{b}_i^2 = \mathbb{E}_g \|x_i f(g^{(i)}) - f(g)x_i\|_2^2.$$

Then $\hat{a}_i^2 \leq \frac{4}{3} \sum_{j>i} \kappa(c_i, c_j)^2$, $\hat{b}_i^2 \leq \frac{4}{3} \sum_{j \neq i} \kappa(x_i, c_j)^2$, and

$$\log D \geq \sum_{i=1}^N \left[\frac{(\Delta_i^2 - \hat{a}_i^2)_+}{3} - \frac{\hat{b}_i^2}{2 \ln 2} - G(\hat{a}_i^2) \right].$$

In particular $\Delta_i \geq 49/100$ and $\hat{a}_i, \hat{b}_i \leq 1/100$ for all i give $\log D \geq N/16$.

Proof. The ordered-product errors. For h unitary, $\|(I - E_j)h\|_2^2 = \frac{1}{3} \kappa(h, c_j)^2$ for the average E_j of conjugation by the powers of c_j . Moving c_i through the suffix $c_{i+1}^{g_{i+1}} \cdots c_N^{g_N}$ and averaging over its exponents gives $\hat{a}_i^2 = 2 - 2 \operatorname{Re} \langle c_i, E_N \cdots E_{i+1} c_i \rangle$, and Lemma 4.1 gives the first bound. For the second, $T_j(Z) = \frac{1}{3} \sum_k c_j^{-k} Z c_j^{s_j k}$ with $s_i = -1$ and $s_j = 1$ otherwise are orthogonal projections, $T_i(x_i) = x_i$ by the local relation, $\|(I - T_j)x_i\|_2^2 = \frac{1}{3} \kappa(x_i, c_j)^2$ for $j \neq i$, and $\hat{b}_i^2 = 2 - 2 \operatorname{Re} \langle x_i, T_N \cdots T_1 x_i \rangle$.

Embedding. Let $V\xi = 3^{-N/2} \sum_g f(g)\xi \otimes |g\rangle$, let C_i shift the i -th coordinate of $\ell^2(\mathbb{Z}_3^N)$ by -1 , let F_i negate it, and put $X_i = x_i \otimes F_i$. The C_i commute, $X_i C_i X_i = C_i^{-1}$, $X_i C_j X_i = C_j$ for $j \neq i$, and $D^{-1/2} \|C_i V - V c_i\|_{HS} = \hat{a}_i$, $D^{-1/2} \|X_i V - V x_i\|_{HS} = \hat{b}_i$.

Entropy. Let $\rho = VV^*/D$ and let $\mathcal{T}_i$ average conjugation by the powers of C_i . The twirl moves weight at most $\min\{\frac{2}{3}\hat{a}_i^2, \frac{2}{3}\}$ outside the range of VV^* , into a block of rank at most $2D$, so $S(\mathcal{T}_i \rho) \leq \log D + G(\hat{a}_i^2)$; as in Lemma 4.2 the twirls commute, and the increment of each twirl, the Holevo quantity of the conjugates of the current state, can only decrease under the earlier twirls by data processing for the Holevo quantity, so $\bar{\rho} = \mathcal{T}_1 \cdots \mathcal{T}_N \rho$ has $S(\bar{\rho}) \leq \log D + \sum_i G(\hat{a}_i^2)$.

Sectors. Let p be the joint distribution of the eigenvalue labels $Z_i \in \{1, \omega, \omega^2\}$ of the C_i , the same for ρ and $\bar{\rho}$. Then $S(\bar{\rho}) \geq H(p) \geq \sum_i H(Z_i | Z_{-i})$. Let Q_a and P_b be the spectral projections of c_i and C_i . The numbers $r_{ab} = D^{-1} \|P_b V Q_a\|_{HS}^2$ form a joint distribution whose marginals are the spectral distribution of c_i in the normalized trace and that of C_i in ρ . Distinct cube roots of unity are at squared distance 3, so $\hat{a}_i^2 = 3 \operatorname{Pr}(a \neq b)$ and $\Delta_i^2 = 3 \operatorname{Pr}(a \neq 1)$, whence $s_i := \operatorname{Pr}(Z_i \neq 1) \geq (\Delta_i^2 - \hat{a}_i^2)_+/3$. Next, the normalized vectorizations of V and of $X_i V x_i^*$ are at squared distance $\hat{b}_i^2$, so their real overlap is $1 - \hat{b}_i^2/2$. Measuring the joint spectral projections of the C_j gives p from the first and its flip $p^{(i)}$ in coordinate i from the second, because X_i exchanges ω and ω^2 in coordinate i and fixes the other labels; by the Cauchy-Schwarz inequality outcome by outcome, $\sum_z \sqrt{p(z)p(z^{(i)})} \geq 1 - \hat{b}_i^2/2$. For $0 \leq t \leq 1$ one has $1 - h_2(t) \leq (1 - 2\sqrt{t(1-t)})/\ln 2$: with $y = 2t - 1$ the difference of $1 - \sqrt{1-y^2}$ and $\ln 2(1 - h_2(t))$ vanishes with its derivative at $y = 0$ and has second derivative $(1-y^2)^{-3/2} - (1-y^2)^{-1} \geq 0$. So for fixed other labels with masses u, w on the two nontrivial values of Z_i , the conditional entropy receives at least $u + w - (u + w - 2\sqrt{uw})/\ln 2$, and summing gives $H(Z_i | Z_{-i}) \geq s_i - \hat{b}_i^2/(2 \ln 2)$. The numbers in the last claim give $\frac{0.49^2 - 10^{-4}}{3} - \frac{10^{-4}}{2 \ln 2} - G(10^{-4}) > \frac{1}{16}$. $\square$

The theorem isolates the row quantities $\hat{a}_i, \hat{b}_i$ that control the dimension estimate used below.

4.3 From an approximate presentation to dimension

Call $\phi : \{a, b, \alpha\} \rightarrow U(D)$ an e -representation if $\|\phi(r_i) - I\|_2 \leq e$ for $i = 1, \dots, 5$, where ϕ is extended to the free group. If w is null in P with area A , then $\|\phi(w) - I\|_2 \leq Ae$, because w is freely a product of A conjugates of relators and the norm is unitarily invariant.

Proposition 4.5. *Let ϕ be an e -representation with $\|\phi(\gamma) - I\|_2 \geq 1/2$, and let $N \geq 1$ satisfy $750\sqrt{N} A_N e \leq 1$, where $A_N = \max\{1, A^*(3N)\}$. Then $\log D \geq N/16$.*

Proof. Put $A_i = \phi(\sigma_{3(i-1)})$ and $B_i = \phi(\sigma_{3(i-1)+1})$ for $1 \leq i \leq N$; all indices are at most $3N - 2$. In H_3 each pair generates a copy of S_3 on $\{(1, 3i - 2), (1, 3i - 1), (1, 3i)\}$; its involution and braid relations are conjugates of r_1 and r_2 , with defect at most e . Lemma 4.3 gives exact local pairs $x_i = u_i(s)$ and $c_i = u_i(st)$ with $\|x_i - A_i\|_2 \leq e$ and $\|c_i - A_i B_i\|_2 \leq 9e$, and $A_i B_i = \phi(a)^{3(i-1)} \phi(\gamma) \phi(a)^{-3(i-1)}$, so $\Delta_i \geq \frac{1}{2} - 9e$. Pairs with different i have indices differing by at least two, so four and two far-commutator interchanges give $\kappa(c_i, c_j) \leq 4A_N e + 36e \leq 40A_N e$ and $\kappa(x_i, c_j) \leq 2A_N e + 20e \leq 22A_N e$. By Theorem 4.4 and the concavity of G , $\log D \geq N[(\bar{\Delta}^2 - \bar{a})/3 - \bar{b}/(2 \ln 2) - G(\bar{a})]$ with $\bar{a} = N^{-1} \sum_i \hat{a}_i^2 \leq \frac{2 \cdot 40^2}{3 \cdot 750^2}$, $\bar{b} = N^{-1} \sum_i \hat{b}_i^2 \leq \frac{4 \cdot 22^2}{3 \cdot 750^2}$ and $\bar{\Delta}^2 \geq (\frac{1}{2} - \frac{9}{750})^2$. Exact rational arithmetic, using $\ln 2 > 69/100$ and $h_2(q) \leq q \log(1/q) + q/\ln 2$, gives a margin of at least $141043/1397250000$ above $1/16$. Explicitly, put $a = 2 \cdot 40^2/(3 \cdot 750^2)$, $b = 4 \cdot 22^2/(3 \cdot 750^2)$ and $q = 2a/3 = 64/50625$. The integer inequality $64^3 2^{29} \geq 50625^3$ gives $\log(1/q) \leq 29/3$, and the resulting rational lower margin is

$$\frac{(488/1000)^2 - a}{3} - \frac{b}{2(69/100)} - \left(\frac{32}{3} + \frac{100}{69}\right)q - \frac{1}{16} = \frac{141043}{1397250000} > 0.$$

□

If instead $\|\phi(\gamma) - I\|_2 \geq \Delta$ with $0 < \Delta \leq 1$, the same argument with 750 replaced by a constant depending only on Δ gives $\log D \geq c_\Delta N$ with $c_\Delta > 0$. With $A^*(M) \leq CM^p$ the condition reads $N^{p+1/2}e = O(1)$, which gives $\log D \geq ce^{-2/(2p+1)}$: the exponent is $2/(2p+1)$ in $1/e$, or $1/(2p+1)$ in the squared defect $1/e^2$. For $p = \log 107$ this is the exponent $2\beta_H$ of Theorem 1.4.

5 Lower bounds

Proposition 5.1. *Let ϕ assign permutations of a finite set Ω to a, b, α , with $d_n(\phi(r_i), \text{id}) \leq \delta$ for all i and $d_n(\phi(\gamma), \text{id}) \geq 1/2$, and let $N \geq 1$ satisfy $10^8 N A_N \delta \leq 1$, where $A_N = \max\{1, A^*(3N)\}$. Then $\log |\Omega| \geq N/16$.*

Proof. Use the pairs $A_i = \phi(\sigma_{3(i-1)})$, $B_i = \phi(\sigma_{3(i-1)+1})$, $1 \leq i \leq N$, as permutation matrices, with $\|P - I\|_2^2 = 2d_n(P, \text{id})$. Their S_3 relators have $d_n \leq \delta$, hence $\|\cdot\|_2$ -defect at most $e = \sqrt{2\delta}$, and Lemma 4.3 gives exact representations u_i within $10e$ of the normal forms. Commuting two normal forms from different pairs takes at most nine far-commutator interchanges, so by transport in measure their commutator has $d_n \leq 9A_N \delta$ and $\|\cdot\|_2 \leq 3\sqrt{2A_N \delta}$; rounding adds at most $40e$, so $\eta \leq 43\sqrt{2A_N \delta}$. The element $u_i(st)$ lies within $9e$ of a conjugate of $\phi(\gamma)$, whose $\|\cdot\|_2$ -displacement is at least 1, so $\Delta \geq 1 - 9\sqrt{2\delta} \geq \frac{1}{2}$. Then $t^2 = 2N\eta^2 \leq 7396N A_N \delta < 10^{-4}$, and Lemma 4.2 gives $\log |\Omega| \geq N[(49/100)^2/3 - h_2(10^{-4}) - 10^{-4} \log 5] \geq N/16$. □

With $A^*(M) \leq CM^p$ the condition reads $N^{p+1}\delta = O(1)$, which gives the sofic exponent $1/(p+1)$.

Proof of Theorem 1.1. Let $f : E_H \rightarrow \text{Sym}_n$ be a $(1 - r^{-1})$ -expansive r^{-1} -morphism with $r \geq 10$, and regard permutations as unitary matrices. For permutation matrices, $\|P - P'\|_2^2 = 2d_n(P, P')$, and d_n is bi-invariant. Put $\phi(s) = f(s)$ for $s \in \{a, b, \alpha\}$ and extend ϕ to the free group. Since $s \cdot s^{-1} = 1$ in the chunk, $d_n(f(s)^{-1}, f(s^{-1})) \leq r^{-1}$. For a relator $s_1 \cdots s_l$ with prefixes p_k , the chunk contains $p_k s_{k+1} = p_{k+1}$ and $p_l = 1$, so $l - 1$ merges and at most l inverse replacements give $d_n(\phi(r_i), \text{id}) \leq (2l - 1)/r \leq 23/r$, that is $\|\phi(r_i) - I\|_2 \leq \sqrt{46/r}$.

In the same way $d_n(\phi(\gamma), f(\gamma)) \leq 4/r$, while expansiveness gives $d_n(f(\gamma), \text{id}) \geq 1 - 1/r$ because $\gamma \neq 1$; so $d_n(\phi(\gamma), \text{id}) \geq 1 - 5/r \geq 1/2$. Apply Proposition 5.1 with $\delta = 23/r$ and $A^*(3N) \leq 135(3N)^{\log 107}$ (Theorem 3.2): the condition $10^8 N \cdot 135(3N)^{\log 107} \delta \leq 1$ holds for $N = \lfloor (r/(23 \cdot 10^8 \cdot 135 \cdot 3^{\log 107}))^{1/(\log 107+1)} \rfloor$. Therefore, whenever $N \geq 1$ (which forces $r \geq 10$),

$$\log \sigma_{E_H}(r) \geq \frac{N}{16} \geq c_S r^{1/(\log 107+1)} - \frac{1}{16},$$

with $c_S = (23 \cdot 10^8 \cdot 135 \cdot 3^{\log 107})^{-1/(\log 107+1)}/16 > 1/1280$, and for smaller r the right side is negative. For unitary approximations Proposition 4.5 gives the corresponding bound: by Theorem 3.2, with $K' = 750 \cdot 135 \cdot 3^{\log 107}$ and $N = \lfloor (K'e)^{-2\beta_H} \rfloor$, every e -representation of P on $\mathbb{C}^D$ with $\|\phi(\gamma) - I\|_2 \geq 1/2$ has $\log D \geq (K'e)^{-2\beta_H}/16 - 1/16$, where $\beta_H = 1/(2 \log 107 + 1)$; this is Theorem 1.4. The upper bound $\log \sigma_{E_H}(r) = O(\sqrt{r} \log r)$ is Proposition 6.2. $\square$

6 Upper bounds

6.1 Permutation models

The upper bounds use permutation models that follow the quotient $\mathbb{Z}^2$ on a torus of clock positions and fail at a single corner. Fix integers $M \geq 2$ and $s \geq 2$, put $\ell = 2M + 2$ and $L = 3\ell$, and let $\Omega_{M,s} = (\mathbb{Z}/M\mathbb{Z})^2 \times [s]^L$. For $2 \leq r \leq L$ let $c_r = (1 \ r \ r-1 \ \cdots \ 2) \in \text{Sym}(L)$ be the reverse cycle and $\tau = (1 \ 2)$; products of permutations are read from left to right, and a permutation π of the coordinates acts on colorings by $(x \cdot \pi)(t) = x(t\pi^{-1})$. For $0 \leq i, j < M$ put $A_{i,j} = c_{\ell-i-j}$ and, for $j < M - 1$, $B_{i,j} = c_{2\ell-j}$. On the top row put $B_{i,M-1} = D_i$, where

$$D_0 = c_{2\ell-M+1}, \quad D_{i+1} = c_{\ell-i-(M-1)}^{-1} \tau D_i c_{\ell-i} \quad (0 \leq i < M-1). \quad (2)$$

Define permutations of $\Omega_{M,s}$, with clock coordinates modulo M , by

$$(i, j, x) \cdot a = (i+1, j, x \cdot A_{i,j}), \quad (i, j, x) \cdot b = (i, j+1, x \cdot B_{i,j}), \quad (i, j, x) \cdot \alpha = (i, j, x \cdot \tau).$$

Proposition 6.1 (Corner models). *In these permutations r_1, r_2, r_3, r_5 hold exactly and r_4 fails only at the clock $(M-1, M-1)$, where it acts as the product $R_M = (1 \ M+4) \prod_{t=5}^{M+3} (t \ t+M)$ of M disjoint transpositions of the colours; so every relator moves at most a fraction M^{-2} of $\Omega_{M,s}$, and γ moves a fraction $1 - s^{-2}$. For $M \geq 5$ the far commutator $C_{2M-1} = [\alpha, a^{2M-1} \alpha a^{1-2M}]$ moves a fraction at least $(1 - s^{-2})(M-4)(M-3)/(2M^2)$. A word of length at most t that represents a nonidentity element of H_3 fixes at most a fraction $s^{-1} + 4t/M$ of $\Omega_{M,s}$ when $M > 2t$.*

Proof. Cycle identities. For $r \geq 4$, $c_r \tau c_r^{-1} = (2 \ 3)$ and $c_r (2 \ 3) c_r^{-1} = (3 \ 4)$, and for $3 \leq r < t$

$$c_r c_t c_{r-1}^{-1} c_t^{-1} = \tau; \quad (3)$$

both sides exchange 1 and 2 and fix every other point. By induction on i , the recursion (2) gives $D_i(t) = t - 1$ for $2 \leq t \leq M + 3 - i$: this holds for the reverse cycle D_0 , and a point $t \leq M + 2 - i$ passes the four factors of D_{i+1} as $t \mapsto t + 1 \mapsto t + 1 \mapsto t \mapsto t - 1$. Since $M + 3 - i \geq 4$, $D_i^{-1}(1) = 2$ and $D_i^{-1}(2) = 3$, so $D_i \tau D_i^{-1} = (2 \ 3)$.

Relators. At every clock the colour permutation of $a\alpha a^{-1}$ is $(2 \ 3)$, that of $a^2 \alpha a^{-2}$ is $(3 \ 4)$, and that of $b\alpha b^{-1}$ is $(2 \ 3)$, on the top row by the property of D_i . Hence r_1, r_2, r_3, r_5 act trivially everywhere, and γ acts at every clock as a three-cycle of coordinates, which fixes a fraction s^{-2} of the colorings. At the clock (i, j) the colour permutation of r_4 is $A_{i,j} B_{i+1,j} A_{i,j+1}^{-1} B_{i,j}^{-1} \tau$. For $j < M - 1$ it is trivial by (3) with $r = \ell - i - j$ and $t = 2\ell - j$, also for $i = M - 1$ because $B_{i,j}$ does not depend on i off the top row. For $j = M - 1$ and

$i < M - 1$ it is trivial by (2). Only the clock $(M - 1, M - 1)$ remains. There the colour permutation is $c_4 D_0 c_{M+3}^{-1} D_{M-1}^{-1} \tau$. With $E_r = c_r^{-1} \tau = (2 \ 3 \ \dots \ r)$ the recursion unrolls to $D_{M-1} = E_5 E_6 \dots E_{M+3} D_0 c_{2M+2} c_{2M+1} \dots c_{M+4}$, and $D_0 c_r^{-1} D_0^{-1} = E_{r+1}$ for $r \leq 3M + 4$, so the corner permutation is $c_4 H G^{-1} \tau$ with $H = E_{M+4} \dots E_{2M+3}$ and $G = E_5 \dots E_{M+3}$. Following the points, H sends $t \mapsto t + M$ for $2 \leq t \leq M + 3$ and $t \mapsto 2M + 5 - t$ for $M + 4 \leq t \leq 2M + 3$, and G sends $t \mapsto t + M - 1$ for $2 \leq t \leq 4$ and $t \mapsto M + 5 - t$ for $5 \leq t \leq M + 3$, both fixing all other points. Hence $HG^{-1} = (2 \ 3 \ 4 \ M+4) \prod_{t=5}^{M+3} (t \ t+M)$, and $c_4 H G^{-1} \tau = R_M$.

Far commutators. Let $k = 2M - 1$, fix a clock (i, j) with $i + j \geq M + 3$ (so $i \geq 4$), and put $F = \ell - j$. The colour permutation of a^k read from this clock is $P = \prod_{t=0}^{k-1} c_{F - ((i+t) \bmod M)}$, and that of σ_k is $P \tau P^{-1}$, which moves $P^{-1}(1)$ and $P^{-1}(2)$. With $d_r = c_r^{-1} = (1 \ 2 \ \dots \ r)$,

$$x \cdot (d_r d_{r+1} \dots d_t) = \begin{cases} x + t - r + 1, & x < r, \\ t - x + 1, & r \leq x \leq t, \\ x, & x > t. \end{cases}$$

The factors of P^{-1} come in three blocks, with cycle lengths $F - i + 2, \dots, F$, then $F - M + 1, \dots, F$, then $F - M + 1, \dots, F - i$. The first block sends $(1, 2)$ to $(i, i + 1)$; since $i \geq F - M + 1$, the second sends $(i, i + 1)$ to $(F - i + 1, F - i)$; the third fixes $F - i + 1$ and sends $F - i$ to 1. So σ_k acts as the transposition $(1 \ 2M + 3 - i - j)$, whose second point is at least 5, and C_k acts as a three-cycle. There are $(M - 4)(M - 3)/2$ such clocks.

Short words. Let $\mathcal{S}_{i,j}$ consist of the first $\ell - i - j$, $\ell + i$ and $\ell + j$ points of rays 1, 2, 3, ordered ray by ray; it has L points. When the clock step does not wrap, the generator a of H_3 maps $\mathcal{S}_{i,j}$ onto $\mathcal{S}_{i+1,j}$ and induces $c_{\ell-i-j}$ on the ordered coordinates; likewise b induces $c_{2\ell-j}$ and α induces τ . So on the clocks at distance at least t from the lines where the clock wraps, a fraction at least $1 - 4t/M$ when $M > 2t$, a word of length at most t acts through the element of H_3 it represents. If that element has nonzero translation it fixes no state. If it is nontrivial with zero translation, it moves only points among the first $t + 2$ of each ray, which lie in $\mathcal{S}_{i,j}$, so it acts as a nonidentity permutation of the coordinates and fixes at most a fraction s^{-1} of the colorings. Its fixed-point fraction is therefore at most $s^{-1} + 4t/M$. $\square$

Proposition 6.2 (Upper bound for every fixed chunk). *For every finite chunk $E \subset H_3$, $\log \sigma_E(r) = O_E(\sqrt{r} \log r)$. In particular this holds for E_H .*

Proof. Fix representative words for the elements of E , with the empty word representing 1, and let t bound the lengths of their pairwise differences. The finitely many chunk relations have some maximum area B_E in Johnson's presentation. Take $s = M$ and the diagonal action on two independent copies of $\Omega_{M,M}$. For $M > 4t$, the short-word estimate in Proposition 6.1 bounds the fixed-point fraction of every nonidentity difference by $((1 + 8t)/M)^2$ in the product: the fractions square. Each relator has defect at most $2/M^2$ in the product, so transport in measure bounds every chunk-relation defect by $2B_E/M^2$. Choosing M at least a sufficiently large E -dependent multiple of $\sqrt{r}$ gives an r^{-1} -morphism that is $(1 - r^{-1})$ -expansive. Its degree is $|\Omega_{M,M}|^2 = M^{4+2(6M+6)}$, whose logarithm is $O(M \log M)$, proving the claim. $\square$

6.2 Unitary models

In a permutation model a relator that fails costs the fraction of points it moves. In a unitary model it costs a sum of squared angles, so a single defect can be spread thin. Doing this for the corner models gives unitary models whose relator defect has squared norm of order M^{-3} , not M^{-2} , at the same size. Throughout, products are read from left to right, the left factor acting first, as in Section 6, and $e_1 = (1, 0)$, $e_2 = (0, 1)$.

One-particle models. No generator moves the last $2M+2$ of the $6M+6$ colour coordinates. Use the reduced colour coordinates $1, \dots, L$ with $L = 4M+4$, and read each link permutation A_c, B_c and τ of Section 6 as a permutation matrix on $\mathbb{C}^L$. A one-particle model replaces some of these link matrices by unitaries. For a word w and a clock c the holonomy $W_c(w) \in U(L)$ is the product of the link matrices along the path of w from c , with inverse matrices for inverse letters. For a two-element set $S = \{x, y\}$ let Q_S be the projector onto $(e_x - e_y)/\sqrt{2}$; then $\exp(i\pi Q_S)$ is the transposition $(x\ y)$, and $\exp(i\vartheta Q_S)$ is a phase rotation in the plane of S .

Fermionic quantization. For $W \in U(L)$ let $\Lambda(W)$ act on the exterior algebra $\Lambda(\mathbb{C}^L)$, of dimension 2^L , by $v_1 \wedge \dots \wedge v_m \mapsto Wv_1 \wedge \dots \wedge Wv_m$. It is multiplicative on all of $U(L)$, and $\text{Tr } \Lambda(W) = \det(I + W)$, since in degree m its eigenvalues are the products of m eigenvalues of W . For $k \geq 1$ put $\Lambda_k = \Lambda^{\otimes k}$, so that $2^{-kL} \text{Tr } \Lambda_k(W) = \det((I + W)/2)^k$. The quantized model acts on $\mathcal{H} = \mathbb{C}^{M^2} \otimes \Lambda(\mathbb{C}^L)^{\otimes k}$, with normalized trace $\tau_{\mathcal{H}}$, distinct from the transposition $\tau = (1\ 2)$. A letter moves the clock c to $c + \pi(s)$ and applies Λ_k of its link matrix at c . These right-action operators compose in reverse order; taking their adjoints defines the usual free-group representation ρ on a, b, α . A word whose translation is nonzero modulo M has $\tau_{\mathcal{H}}(\rho(w)) = 0$, and a word that closes has

$$\|\rho(w) - I\|_2^2 = \mathbb{E}_c \left[2 - 2 \text{Re} \det((I + W_c(w))/2)^k \right],$$

with $\mathbb{E}_c$ the average over clocks. For a three-cycle σ of coordinates $\det((I + \sigma)/2) = \frac{1}{4}$.

Lemma 6.3 (Rotating the links). *In a corner model, replace every link A_c by $U_c A_c$ and every link B_c by $V_c B_c$, where U_c commutes with $(2\ 3)$ and $(3\ 4)$, and V_c commutes with $(2\ 3)$. Then at every clock the holonomies of $\alpha, a\alpha a^{-1}, a^2\alpha a^{-2}$ and $b\alpha b^{-1}$ are $(1\ 2), (2\ 3), (3\ 4)$ and $(2\ 3)$, so r_1, r_2, r_3, r_5 have identity holonomy everywhere and γ acts as a three-cycle.*

Proof. By the proof of Proposition 6.1, $A_c \tau A_c^{-1} = (2\ 3)$, $A_c(2\ 3)A_c^{-1} = (3\ 4)$ and $B_c \tau B_c^{-1} = (2\ 3)$ at every clock. So the holonomy of $a\alpha a^{-1}$ at c is $U_c(2\ 3)U_c^{-1} = (2\ 3)$, that of $a^2\alpha a^{-2}$ is $U_c A_c U_{c+e_1}(2\ 3)U_{c+e_1}^{-1}A_c^{-1}U_c^{-1} = U_c(3\ 4)U_c^{-1} = (3\ 4)$, and that of $b\alpha b^{-1}$ is $V_c(2\ 3)V_c^{-1} = (2\ 3)$. The relators r_1, r_2, r_3, r_5 are products of these conjugates, as in the proof of Proposition 6.1. $\square$

A plaquette q with base v_0 has the vertices $v_0, v_1 = v_0 + e_1, v_2 = v_0 + e_2, v_0 + e_1 + e_2$; its bottom and top edges are the a -links at v_0 and v_2 , and its left and right edges the b -links at v_0 and v_1 . Its r_4 holonomy is $W_q = A_{v_0} B_{v_1} A_{v_2}^{-1} B_{v_0}^{-1} \tau$. A *region* consists of a set R of plaquettes, a source $q_0 \in R$, a sign $s_R = \pm 1$, and a two-element label $\text{lab}(v) \subset \{1, \dots, L\}$ at every vertex of every plaquette of R . An edge is *interior* if it lies on two plaquettes of R .

Lemma 6.4 (Spreading a defect). *Take a permutation model satisfying the local identities $A_c \tau A_c^{-1} = (2\ 3)$, $A_c(2\ 3)A_c^{-1} = (3\ 4)$ and $B_c \tau B_c^{-1} = (2\ 3)$ of Lemma 6.3 at every clock. Suppose W_q is the identity except at finitely many sources, where it is the product of the transpositions given by the base labels of the regions with that source. Let finitely many regions satisfy:*

- (R1) *for each region R , every edge of a plaquette $q \in R$ other than its own source $q_0(R)$, from v to w , carries $\text{lab}_R(v)$ onto $\text{lab}_R(w)$ under its link permutation, even if q is another region's source;*
- (R2) *every base label avoids $\{1, 2\}$, and the label at the initial vertex of every interior edge avoids $\{2, 3, 4\}$;*
- (R3) *each region is connected through its interior edges, and the only interior edge of its source is the bottom edge;*

(R4) labels of different regions are disjoint wherever both are defined; on an interior link carrying a region's rotation, its label is disjoint from any additional modification of that link.

Here the link permutations belong to the base model, including any preliminary exact moves. Potentials vanish on boundary links; no disjointness from an additional modification is required for a region on a link where that region carries no rotation. Let $F_{R,q}$ be real numbers with $\sum_{q \in R} F_{R,q} = s_R \pi$ for each region. Then there are potentials φ_R on the interior edges of R such that the model with $U_c = \prod_R \exp(i\varphi_R(a\text{-edge at } c) Q_{\text{lab}_R(c)})$ and $V_c = \prod_R \exp(i\varphi_R(b\text{-edge at } c) Q_{\text{lab}_R(c)})$ still has r_1, r_2, r_3, r_5 exact, and its r_4 holonomy at every plaquette q is $\prod_{R \ni q} \exp(iF_{R,q} Q_{\text{lab}_R(v_0)})$, with v_0 the base of q .

Proof. Put $\varphi_R = 0$ off the interior edges and $(\text{curl } \varphi)(q) = \varphi(\text{bottom}) + \varphi(\text{right}) - \varphi(\text{top}) - \varphi(\text{left})$. An interior edge enters the curls of its two plaquettes with opposite signs, so the curls over R sum to zero. Choose a spanning tree of the plaquettes of R joined through interior edges, rooted at the source, and fix the potential of each tree edge, from the leaves inward, so that $\text{curl } \varphi_R = F_R - s_R \pi \delta_{q_0}$ at the plaquette below it; since both sides sum to zero, the equation then also holds at the root. By (R2) and (R4) all rotations on a link commute with the transpositions of Lemma 6.3, so r_1, r_2, r_3, r_5 stay exact.

For the r_4 holonomy, the unmodified identity $A_{v_0} B_{v_1} A_{v_2}^{-1} B_{v_0}^{-1} = W_q \tau$ gives

$$W'_q = U_{v_0} (A_{v_0} V_{v_1} A_{v_0}^{-1}) W_q \tau (B_{v_0} U_{v_2} B_{v_0}^{-1})^{-1} V_{v_0}^{-1} \tau.$$

Conjugation by a link permutation moves the plane of a rotation. For each region separately, at a plaquette other than its own source, (R1) moves the planes $\text{lab}(v_1)$ and $\text{lab}(v_2)$ to $\text{lab}(v_0)$. At that region's own source only its bottom edge carries its potential; potentials of other regions are still treated by their respective cases. By (R4) their planes commute with the source defects of the other regions. So all rotations in W'_q lie in the planes $\text{lab}_R(v_0)$; they avoid $\{1, 2\}$ by (R2), hence commute with τ , and planes of different regions are disjoint by (R4). For region R the angles add up to $(\text{curl } \varphi_R)(q)$, and at its source W_q contributes the rotation by $s_R \pi$ in the plane of the source label. The total is $F_{R,q}$. $\square$

Proposition 6.5 (Unitary models). *For every $M \geq 48$ the model above with $k = 2$ and suitable rotations of the links gives unitaries $\rho(a), \rho(b), \rho(\alpha)$ on a space of dimension $M^2 2^{8M+8}$ in which r_1, r_2, r_3, r_5 act as the identity, $\|\rho(r_4) - I\|_2^2 \leq 14.26 M^{-3}$ and $\|\rho(\gamma) - I\|_2^2 = 15/8$.*

Proof. Two exact moves. Let $c^* = (M-1, M-1)$, where the r_4 holonomy is R_M (Proposition 6.1). Replace A_{c^*} by $\eta_1 A_{c^*}$ and $B_{(0, M-2)}$ by $\eta_2 B_{(0, M-2)}$, with $\eta_1 = (1 \ M+4)$ and $\eta_2 = (5 \ M+5)$. These satisfy the hypotheses of Lemma 6.3, so r_1, r_2, r_3, r_5 stay exact. The modified links lie on three plaquettes. At c^* the holonomy becomes $\eta_1 R_M = \prod_{t=5}^{M+3} (t \ t+M)$. At $(0, M-2)$, where it was the identity, it becomes $\tau \eta_2 \tau = \eta_2$. At $(M-1, M-2)$ it becomes $(A \eta_2 A^{-1}) \tau (B \eta_1 B^{-1}) \tau$ with $A = A_{(M-1, M-2)} = c_5$ and $B = B_{(M-1, M-2)} = c_{3M+6}$. Here $A \eta_2 A^{-1} = (1 \ M+5)$ and $B \eta_1 B^{-1} = (2 \ M+5)$, so $\tau (B \eta_1 B^{-1}) \tau = (1 \ M+5)$ and the holonomy stays the identity. Use these modified permutations as the base model in Lemma 6.4.

The regions. For $5 \leq t \leq M+3$ let R_t have source c^* and sign $(-1)^{t-5}$, and consist of c^* , the plaquettes $(M-1, j)$ with $1 \leq j \leq M-2$, and the plaquettes (i, j) with $0 \leq i \leq M-3$, $0 \leq j \leq M-2$ and $i+j \leq t+M-8$; label the vertex (i, j) by $\{t+M-1-i-j, t+2M-1-j\}$ for $0 \leq i \leq M-2$ and $(M-1, j)$ by $\{t+M-1-j, t+2M-1-j\}$. Let T have source $(0, M-2)$ and sign $(-1)^{M-1}$, and consist of $(0, M-2)$ and the plaquettes (i, j) with $i, j \geq 0$ and $i+j \leq M-3$, with labels $\{M+3-i-j, 2M+3-j\}$. The source labels are the defect pairs $\{t, t+M\}$ at c^* and $\{5, M+5\}$ at $(0, M-2)$.

These regions satisfy (R1)–(R4). Off the top row $B_{i,j} = c_{4M+4-j}$ lowers every label coordinate by one, which is the change of the labels from j to $j+1$. In the triangles

$A_{i,j} = c_{2M+2-i-j}$ lowers the first label coordinate, which is at most $2M + 2 - i - j$ since $t \leq M + 3$, and fixes the second, which exceeds $2M + 2 - i - j$ since $t \geq 5$; this is the change from i to $i + 1$, and the same holds in T . On the column $i = M - 1$, $A_{M-1,j} = c_{M+3-j}$ fixes both label coordinates, which are at least $t + M - 1 - j \geq M + 4 - j$, and the labels at $(M - 1, j)$ and $(0, j)$ agree; this is the only crossing of a seam, since the rows stay below $M - 1$ and c^* is attached only through its bottom edge. Every base label and every label at the initial vertex of an interior edge has both coordinates at least 5. Two labels of bulk regions at one vertex differ in t by at most $M - 2$, so they are disjoint; a bulk label meets a label of T only if $t \in \{4, 4 - M - i, M + 4 + i\}$, which is impossible; and η_1, η_2 are disjoint from the labels $\{t, t + M\}$ and $\{t + 1, t + M + 1\}$ on the links where those regions carry rotations. At $(0, M - 2)$ the label of T equals the support of η_2 , but that b -link is a boundary link of T and carries no T -rotation. Each region is connected through interior edges: the triangle is a staircase, the column is glued to it through the b -edges at $(0, j)$, and each source through its bottom edge. Finally $R_t \subset R_{t+1}$, and every region has at least $m_M = (M - 1)(M - 2)/2 + 1$ plaquettes.

The defect. Take $F_{R,q} = s_R \pi / |R|$ on each region and $k = 2$. By Lemma 6.4 the r_4 holonomy at q is a product of phase rotations by angles $F_{R,q}$ in disjoint planes, so $\det((I + W'_q)/2) = \prod_R (1 + e^{iF_{R,q}})/2 = e^{i\Theta_q} C_q$ with $\Theta_q = \frac{1}{2} \sum_R F_{R,q}$ and $C_q = \prod_R \cos(F_{R,q}/2) \in [0, 1]$. Using $1 - C^k \leq k(1 - C)$, $1 - \prod_R c_R \leq \sum_R (1 - c_R)$ and $1 - \cos x \leq x^2/2$,

$$2 - 2 \operatorname{Re}(e^{ik\Theta_q} C_q^k) = 2(1 - C_q^k) + 2C_q^k(1 - \cos k\Theta_q) \leq \frac{k}{4} \sum_R F_{R,q}^2 + k^2 \Theta_q^2.$$

Summed over the plaquettes, $\sum_q \sum_R F_{R,q}^2 = \sum_R \pi^2 / |R| \leq M\pi^2 / m_M$. The bulk regions containing q are those with t at least some t_q ; their signs alternate and their magnitudes $\pi / |R_t|$ decrease, so they sum to at most π / m_M in absolute value, and T adds at most π / m_M . Hence $|\Theta_q| \leq \pi / m_M$, and averaging over the M^2 clocks,

$$M^3 e_4^2 := M^3 \|\rho(r_4) - I\|_2^2 \leq \frac{\pi^2 M^2}{2m_M} + \frac{4\pi^2 M^3}{m_M^2} \leq \frac{\pi^2 M^2}{(M - 1)(M - 2)} + \frac{16\pi^2 M^3}{(M - 1)^2 (M - 2)^2},$$

which decreases in M and is below 14.26 at $M = 48$. So $e_4^2 \leq 14.26 M^{-3}$ for $M \geq 48$.

The three-cycle. By Lemma 6.3, γ acts at every clock as a three-cycle σ of coordinates, so $\tau(\rho(\gamma)) = \det((I + \sigma)/2)^2 = 4^{-2}$ and $\|\rho(\gamma) - I\|_2^2 = 2 - 2 \cdot 4^{-2} = 15/8$. $\square$

Proposition 6.6 (Sharp defect within the fixed-region construction). *Fix an integer $k \geq 1$, the clock torus, the regions and their signs in the proof above, and the k -fold exterior-algebra representation. Allow arbitrary real fields $F_{R,q}$ with $\sum_{q \in R} F_{R,q} = s_R \pi$, realized by Lemma 6.4. If $E_{M,k}$ is the infimum of the squared r_4 defect over these fields, then*

$$E_{M,k} = \frac{k\pi^2 \sqrt{2} \ln(1 + \sqrt{2})}{4} M^{-3} (1 + o(1)) \quad (M \rightarrow \infty).$$

This optimizes only these fixed regions and commuting phase planes, not arbitrary unitary models of H_3 .

Proof. Put $S_M = \sum_R |R|^{-1}$ and $m_M = 1 + (M - 1)(M - 2)/2$. Reduce each field angle modulo 2π to $\theta_{R,q} \in [-\pi, \pi]$. Its sum over a region is an odd multiple of π , so $\sum_{q \in R} \theta_{R,q}^2 \geq \pi^2 / |R|$ by Cauchy–Schwarz. The absolute normalized exterior trace at q is $|\prod_{R \ni q} \cos(\theta_{R,q}/2)|^k \leq \exp(-k \sum_{R \ni q} \theta_{R,q}^2 / 8)$. Here $\cos x \leq e^{-x^2/2}$ for $|x| \leq \pi/2$, as follows by differentiating $\log \cos x + x^2/2$. For nonnegative a_q , $\sum_q (1 - e^{-a_q}) \geq 1 - e^{-\sum_q a_q}$. Consequently

$$\frac{2}{M^2} (1 - e^{-k\pi^2 S_M / 8}) \leq E_{M,k} \leq \frac{k\pi^2 S_M}{4M^2} + \frac{k^2 \pi^2}{m_M^2}.$$

The upper bound is the constant field in each region, with the alternating signs and phase estimate in the preceding proof. Directly counting the staircase gives $|R_t| = (M-1)^2 - h(h+1)/2$ for $h = M+3-t \in \{0, \dots, M-2\}$ and $|T| = m_M$. The Riemann sum therefore yields

$$MS_M \longrightarrow \int_0^1 \frac{dx}{1-x^2/2} = \sqrt{2} \ln(1+\sqrt{2}).$$

Since $S_M = O(M^{-1})$ and m_M is of order M^2 , the two bounds have the same leading term. $\square$

Corollary 6.7 (Unitary upper bound). *Let $0 < e \leq e_0 = 14.26^{1/2} \cdot 48^{-3/2}$ and $M = \lceil (14.26/e^2)^{1/3} \rceil$. Johnson's presentation has an e -representation on $\mathbb{C}^D$ with $\|\phi(\gamma) - I\|_2 \geq 1/2$ and $\log D \leq 8M + 8 + 2 \log M$. So such representations exist with $\log D = O(e^{-2/3})$.*

Proof. The choice of M gives $M \geq 48$ and $14.26 M^{-3} \leq e^2$; apply Proposition 6.5. $\square$

Our permutation construction gives the weaker bound in this norm. Read as permutation matrices, the corner models with $s = 2$ have relator defects at most $\sqrt{2}/M$ in $\|\cdot\|_2$, since a permutation moving a fraction f of the points has $\|P - I\|_2^2 = 2f$, and $\|\phi(\gamma) - I\|_2^2 = 3/2$, at $\log D = 6M + 6 + 2 \log M$; this gives only $\log D = O(e^{-1})$. Spreading the defect lowers the exponent of $1/e$ in our upper bounds from 1 to $\frac{2}{3}$; whether optimal permutation models are weaker than optimal unitary ones is open (Section 8). The lower bound of Theorem 1.4 has exponent $2\beta_H \approx 0.138$.

7 The correction

Corollary 7.1. *Theorem 3.15 of [2] is false for $1 \rightarrow \text{FSym}(\mathbb{N}) \rightarrow H_3 \rightarrow \mathbb{Z}^2 \rightarrow 1$ and the chunk E_H . Consequently the conclusion of [2, Example 3.16], that every elementary amenable group has polynomial sofic profile, fails for H_3 .*

Proof. Theorem 3.15 of [2] asserts a symmetric chunk E' of the kernel and a finite symmetric $S \subset \mathbb{Z}^2$ with $\sigma_{E_H}(r) \leq \sigma_{E'}(r) \alpha_{\mathbb{Z}^2, S}(r)$ for all $r > 1$, where $\alpha_{Q, S}(r)$ is the least size of a finite $F \subset Q$ with $|SF \setminus F| < |F|/r$. The kernel $\text{FSym}(\mathbb{N})$ is locally finite, so E' lies in a finite subgroup and $\sigma_{E'}$ is bounded [2, Fact 3.5]; boxes show $\alpha_{\mathbb{Z}^2, S}(r) = O(r^2)$. The resulting bound $\sigma_{E_H}(r) = O(r^2)$ contradicts Theorem 1.1. Since H_3 is elementary amenable, Example 3.16 fails. The contradiction does not need Theorem 3.2: with the certified bound of Theorem 3.1 alone, the proof of Theorem 1.1 gives $\sigma_{E_H}(r) \geq 2^{r^{1/12}/270-1/16}$. $\square$

Where the argument breaks. Theorem 3.15 is stated in [2] as what the explicit construction of Elek and Szabó [3] yields “without any change”. That construction, given a finite $F \subset G$ and $\varepsilon > 0$, chooses a Følner set $A \subset Q$ with $|Ag \setminus A| \leq \varepsilon|A|$ for $g \in F$, a section $\sigma : Q \rightarrow G$, and then an approximation of the kernel on the finite set

$$H_\varepsilon = N \cap (\sigma(A) F \sigma(A)^{-1}),$$

acting on $B \times A$ with B the kernel's approximation space [3, proof of Theorem 1(3)]. The kernel chunk is therefore H_ε , and it depends on $\varepsilon = 1/r$ through A . For H_3 with the section $(m, n) \mapsto a^m b^n$ and $A = [0, M)^2$, it contains $a^m b^n a b^{-n} a^{-m}$, the transposition of $(1, m+n+1)$ and $(1, m+n+2)$, for all $0 \leq m, n < M$: at least $2M-1$ distinct transpositions, which generate the symmetric group of $2M$ points, with M of order r . No chunk independent of r contains them. What the construction does give is the bound of Proposition 7.2 below, in which the kernel chunk grows with r . The upper bound by the isoperimetric profile [2, Proposition 3.14] is not affected, and H_3 , having exponential growth, has isoperimetric profile at least exponential. The proof of [2, Proposition 3.17] for Baumslag–Solitar groups applies Theorem 3.15 in the same way, so it is incomplete as well; we do not know whether its conclusion holds.

A scale-dependent extension bound. The construction of Elek and Szabó, with the accuracy tracked explicitly, gives the following correct form of the extension bound. Let $1 \rightarrow N \rightarrow G \xrightarrow{\pi} Q \rightarrow 1$ be an extension, $s : Q \rightarrow G$ a section with $s(e) = 1$, and put $c(a, g) = s(a) g s(a\pi(g))^{-1} \in N$ for $a \in Q$ and $g \in G$.

Proposition 7.2. *Let $E \subset G$ be a symmetric chunk, $r > 1$, and $A \subset Q$ a finite nonempty set with $|A\pi(g) \setminus A| \leq |A|/(4r)$ for every $g \in E$. Then $K_{A,E} = \{c(a, g) : g \in E, a \in A, a\pi(g) \in A\}$ is a symmetric chunk of N , and*

$$\sigma_E(r) \leq |A| \sigma_{K_{A,E}}(2r).$$

Proof. The set $K_{A,E}$ contains $c(a, 1) = 1$, and $c(a, g)^{-1} = c(a\pi(g), g^{-1})$, so it is symmetric. The cocycle identity $c(a, g) c(a\pi(g), h) = c(a, gh)$ holds in G . Let $f_K : K_{A,E} \rightarrow \text{Sym}_m$ be a $(1 - \frac{1}{2r})$ -expansive $\frac{1}{2r}$ -morphism with $m = \sigma_{K_{A,E}}(2r)$, and on $A \times [m]$ let $f(g)$ send (a, x) to $(a\pi(g), x \cdot f_K(c(a, g)))$ whenever $a\pi(g) \in A$, completed arbitrarily to a permutation; $f(1)$ is the identity. Call a fiber a good for a relation $gh = k$ in E if $a, a\pi(g)$ and $a\pi(k)$ lie in A . Then $c(a, g)c(a\pi(g), h) = c(a, k)$ is a relation of the chunk $K_{A,E}$, so on a good fiber $f(g)f(h)$ and $f(k)$ differ on at most a fraction $\frac{1}{2r}$ of the points, and the bad fibers form a fraction at most $\frac{2}{4r}$. So f is an r^{-1} -morphism. For distinct $g, h \in E$: if $\pi(g) \neq \pi(h)$, then on every fiber a with $a\pi(g), a\pi(h) \in A$ the two images have different quotient coordinates, a fraction at least $1 - \frac{1}{2r}$ of the points; if $\pi(g) = \pi(h)$, then $c(a, g) \neq c(a, h)$ are distinct elements of $K_{A,E}$, so on the fibers with $a\pi(g) \in A$ the images differ on a fraction at least $1 - \frac{1}{2r}$ of each fiber, and in total on at least $(1 - \frac{1}{4r})(1 - \frac{1}{2r}) \geq 1 - \frac{1}{r}$. So f is $(1 - r^{-1})$ -expansive, on $|A| m$ points. $\square$

When the cocycle takes finitely many values on $Q \times E$, for instance for direct products, $K_{A,E}$ lies in a fixed finite set and a bound with a fixed kernel chunk follows, as Theorem 3.15 asserts. For H_3 with boxes $A = [0, M]^2$ and M of order r , K_{A,E_H} lies in the finite group of permutations of the first $2M + O(1)$ points of ray 1, whose regular representation gives $\sigma_{K_{A,E_H}} \leq (2M + O(1))!$; so the proposition gives $\log \sigma_{E_H}(r) = O(r \log r)$, consistent with, and weaker than, Proposition 6.2.

Statement identifiers in [2] refer to the published journal version: Theorem 3.15, Example 3.16 and Proposition 3.17 are on p. 833, and Problem 3.18 is on p. 834. Proposition 3.17 is labelled Example 3.17 in [arXiv:1305.0993v1](#).

8 Open problems

- (1) *The exponents.* The sofic exponent of E_H in r lies between $1/(\log 107 + 1) \approx 1/7.74$ and $1/2$ (up to the logarithm). For unitary approximations with relator defect e in the normalized Hilbert–Schmidt norm, $\log D$ lies between order $e^{-2\beta_H}$, with $2\beta_H \approx 0.138$ (Theorem 1.4), and $O(e^{-2/3})$ (Corollary 6.7). The auxiliary certificate scheme also gives exponent $\log \rho \approx 6.73252$, with $\rho \approx 106.3385$, but the explicit bounds use $\log 107$. These are bounds for one filling scheme, not obstructions to other fillings. Since the area is at least quadratic (Proposition 3.4), the present uniform-area transport and sector argument cannot give more than $1/3$ in r for permutations and $2/5$ in $1/e$ for unitaries. A quadratic Dehn function for H_3 would reach these values: every word in the argument has linear length. Theorem 3.3 gives the polynomial upper bound $O(n^{\log 107 + 4})$, while for Thompson’s group F a quadratic Dehn function is a theorem [13]. Shared fillings, improved aggregate defect estimates, or a stronger dimension argument are possible routes beyond these limits.

- (2) *Permutation versus unitary approximation.* Relator errors add up in measure for permutations and in norm for unitaries, so the two profiles may differ. In common units, a permutation matrix has squared normalized Hilbert–Schmidt defect twice its normalized Hamming defect. At squared defect $\delta = e^2$, our permutation models give $\log D = O(\delta^{-1/2})$ and unitary models $O(\delta^{-1/3})$ (Section 6.2). Different upper constructions do not separate the optimal profiles. A permutation lower exponent strictly above $1/3$ would separate them using this unitary upper bound; the present area method cannot exceed $1/3$. Can a different method prove a separation?
- (3) *Other groups.* Every Houghton group H_m with $m \geq 3$ contains H_3 ; the same method applies to groups with a displaced S_3 structure and polynomially bounded far-commutator area. The proof of [2, Proposition 3.17] for Baumslag–Solitar groups uses the same extension theorem. Its conclusion is already true in every residually finite case: a finite quotient injective on a fixed chunk gives a bounded profile, via its regular permutation action. In particular $BS(1, n)$ is not an open case. For $|n| \geq 2$, its description $\mathbb{Z}[1/n] \rtimes \mathbb{Z}$ proves residual finiteness directly: reduce a nonzero translation modulo an integer coprime to n , then reduce the exponent modulo a multiple of the order of n in that finite ring; a nonzero exponent is separated by a cyclic quotient. The cases $n = \pm 1$ are virtually abelian. Only non-residually-finite Baumslag–Solitar cases remain at issue here.
- (4) *Extensions.* Proposition 7.2 bounds the profile of an extension by the Følner size of the quotient and the profile of a kernel chunk that grows with r through the cocycle of a section; the kernel chunk is fixed when the cocycle takes finitely many values. Which extensions of groups with polynomial profile by amenable quotients have polynomial profile?

A Fillings for the far-commutator bound

Areas are taken in P ; free conjugation and inversion do not change them, and freely $[x, yz] = [x, y] \cdot y[x, z]y^{-1}$. Write $\tau_j = b^j \alpha b^{-j}$, $t = b^{-1}a$, $\rho = a^{-1}b^{-1}ab$, $t_i = a^i t a^{-i}$, $X_n = b^{-n}a^n$, and

$$A(d) = \text{Area}[\sigma_0, \sigma_d], \quad E(i) = \text{Area}[\sigma_i, t], \quad U(j) = \text{Area}(\tau_j \sigma_j^{-1}), \quad B(k) = \text{Area}[\sigma_{-k}, b].$$

Since $b^{-n}(\tau_n \sigma_n^{-1})b^n = [\alpha, X_n]$ freely, $U(n) = \text{Area}[\alpha, X_n]$. Since $[x^{-1}, y] = x^{-1}[y, x]x$, the signs in the definition of A^* do not matter and $A^*(M) = \max_{2 \leq d \leq M} A(d)$. Every relation below with a numerical area is certified in the ancillary files (Section 3).

Base relations. $[\alpha, t]$ is a rotation of r_5^{-1} ; $aba^{-1} = ab$ and $bab^{-1} = \alpha^{-1}a$ are rotations of $r_4^{\pm 1}$; $[\sigma_i, \sigma_{i+2}]$ is a conjugate of r_3 ; all have area 1. The relation $B_0 := \text{Area}[\alpha, \sigma_1 \sigma_0 b] \leq 5$ follows by replacing $b\alpha^{-1}b^{-1}$ with σ_1^{-1} (one r_5), which leaves $\sigma_0 \sigma_1 \sigma_0 \sigma_1^{-1} \sigma_0^{-1} \sigma_1^{-1}$, then three r_1 cells and one r_2 cell.

Lemma A.1 (Transfer and halving). *For $i \geq 1$, $j \geq 1$, $k \geq 2$:*

- (i) $E(i) \leq 2i + 1 + \sum_{d=2}^{i+1} A(d)$, and $E(0) = 1$;
- (ii) $U(j) \leq \sum_{i=0}^{j-1} E(i)$;
- (iii) $B(k) \leq 2k + 5 + \sum_{d=2}^{k-1} A(d)$;
- (iv) $A(k + j) \leq A(k) + 2jB(k) + 2U(j)$.

Proof. (i) By $bab^{-1} = \alpha^{-1}a$, $t_k = t_{k+1}\sigma_{k-1}$ with one cell, so $t = t_i\sigma_{i-2}\cdots\sigma_0\sigma_{-1}$ with i cells, and i more for t^{-1} . Then $[\sigma_i, t_iS] = [\sigma_i, t_i] \cdot t_i[\sigma_i, S]t_i^{-1}$ with $S = \sigma_{i-2}\cdots\sigma_{-1}$; the first factor is $a^i[\alpha, t]a^{-i}$, and the second is a product of conjugates of $[\sigma_i, \sigma_l]$ for $l = i-2, \dots, -1$, of widths $2, \dots, i+1$. (ii) Freely $\tau_{i+1} = b\tau_i b^{-1}$; replace τ_i by σ_i at cost $U(i)$, and use that $b\sigma_i b^{-1}\sigma_{i+1}^{-1} = b[\sigma_i, t]b^{-1}$ freely. (iii) Freely $[\sigma_{-k}, b] = a^{-k}[\alpha, a^k b a^{-k}]a^k$, and $a^k b a^{-k} = \sigma_{k-1}\cdots\sigma_1\sigma_0 b$ with k cells (twice, for the element and its inverse); then move α past $\sigma_{k-1}, \dots, \sigma_2$ and finish with B_0 . (iv) Conjugating by a^{-k} turns $[\sigma_0, \sigma_{k+j}]$ into $[\sigma_{-k}, \sigma_j]$. Replace $\sigma_j^{\pm 1}$ by $\tau_j^{\pm 1}$ ($2U(j)$), move $\sigma_{-k}^{\pm 1}$ across the $2j$ letters $b^{\pm 1}$ ($2jB(k)$), and what remains is $b^j[\sigma_{-k}, \alpha]b^{-j}$, a conjugate of $[\sigma_0, \sigma_k]$. $\square$

The lemma gives, in this order: $A(2) = 1$, $E(0) = U(1) = 1$, $E(1) \leq 4$, $U(2) \leq 5$, $B(2) \leq 9$, $A(3) \leq 21$ ($k = 2, j = 1$), $A(4) \leq 47$ ($k = j = 2$), $E(2) \leq 27$, $U(3) \leq 32$, $B(3) \leq 12$, $A(5) \leq 79$ ($k = 3, j = 2$), $A(6) \leq 157$ ($k = j = 3$) and $E(3) \leq 76$. Part (iv) with $k = 2$ gives $A(m) \leq 2U(m-2) + 18m - 35$ for $m \geq 3$.

The cheap letters. We use $g^h = h^{-1}gh$. Freely, $\psi(t) = b^{-2}a^2 = t\rho t$, $t^b = b^{-2}ab = t\rho$, and $\psi(\rho) = \rho^a \rho^{ba} \rho \rho^b = t^{-1} \rho^b \rho^{b^2} t \cdot \rho \cdot \rho^b$ (from $[x^2, y^2] = (xcx^{-1})(xycy^{-1}x^{-1})c(ycy^{-1})$ with $c = [x, y]$, $x = a^{-1}$, $y = b^{-1}$, and $a = bt$). The one relation needed is

$$\rho^b = R' := t\rho t^{-1} \cdot \rho \cdot t\rho t^{-1}, \quad \text{area} \leq 14.$$

Replace each ρ by α^{ba} (four r_4 cells); freely the relation becomes $\alpha^{bab} = \alpha^{b^2} \alpha^{ba} \alpha^{b^2}$, and after conjugating by $(bab)^{-1}$ it reads $\alpha = \alpha^g \alpha^{g'} \alpha^g$ with $g = ba^{-1}b^{-1}$ and $g' = bab^{-1}a^{-1}b^{-1}$. Six cells of $bab^{-1} = \alpha^{-1}a$ turn this into $\alpha = \alpha^{-1}\sigma_1\alpha \cdot bab^{-1} \cdot \alpha^{-1}\sigma_1\alpha$; one r_5 cell, two r_1 cells and one r_2 cell finish. Also $[\alpha, t]$ has area 1 and $[\alpha, \rho]$ area at most 5: $\rho\alpha^{-1}\rho^{-1}$ becomes $a^{-1}b^{-1}\sigma_2^{-1}ba$ after one r_5 cell, and $b^{-1}\sigma_2^{-1}b \rightarrow \sigma_1^{-1}$ costs $E(1) \leq 4$. Consequently, for a word W in $t^{\pm 1}, \rho^{\pm 1}$, $W^b = W(t\rho, R')$ at area $14 \#_\rho(W)$, and $\psi(W) = W(t\rho, R)$ at area $84 \#_\rho(W)$, where

$$R = t^{-1} \cdot R' \cdot R'(t\rho, R') \cdot t \cdot \rho \cdot R'$$

uses the relation six times per letter ρ ; $|R'| = 7$ and $|R| = 46$.

Lemma A.2 (The doubling lifts). $\psi(\alpha) = \Pi := \sigma_1\sigma_2\sigma_0\sigma_1$ with 6 cells, and $\text{Area } \psi(r_i) \leq 17, 359, 864, 0, 182$ for $i = 1, \dots, 5$. Hence $\text{Area } \psi(w) \leq 864 \text{Area}(w)$ for every null word w .

Proof. The identity for $[x^2, y^2]$ with $x = a$, $y = b$ and four r_4 cells ($[a, b] = \alpha$) give $\psi(\alpha) = \sigma_1 \cdot a(bab^{-1})a^{-1} \cdot \alpha \cdot bab^{-1}$, and two r_5 cells give Π . $\psi(r_4)$ is freely trivial. $\psi(r_1) = \psi(\alpha)^2$: after two conversions, Π^2 costs five cells (σ_1^2 , one r_3 swap, then $\sigma_2^2, \sigma_0^2, \sigma_1^2$). $\psi(r_5) = a^2\psi(\alpha)a^{-2} \cdot b^2\psi(\alpha)^{-1}b^{-2}$: after two conversions, $a^2\Pi a^{-2} = \sigma_3\sigma_4\sigma_2\sigma_3$ freely, and $b^2\sigma_i^{-1}b^{-2} \rightarrow \sigma_{i+2}^{-1}$ costs $E(i) + E(i+1)$; over the letters of Π this is $E(0) + 3E(1) + 3E(2) + E(3) \leq 170$. $\psi(r_3) = [\psi(\alpha), a^4\psi(\alpha)a^{-4}]$: after four conversions it is $[\Pi, \sigma_5\sigma_6\sigma_4\sigma_5]$, which splits freely into sixteen conjugates of far commutators of widths $2, 3, 3, 3, 3, 4, 4, 4, 4, 4, 5, 5, 5, 5, 6$, costing 840. $\psi(r_2) = (\psi(\alpha)a^2\psi(\alpha)a^{-2})^3$: after six conversions it is the positive word $(\Pi\sigma_3\sigma_4\sigma_2\sigma_3)^3$ of length 24 in $\sigma_0, \dots, \sigma_4$, which represents the identity of $\text{Sym}(6)$. Reduce it by the coset procedure: keep the word in the form $h \cdot c_k$ with $c_k = \sigma_4\sigma_3 \cdots \sigma_k$ and h a word in $\sigma_0, \dots, \sigma_3$; absorbing a letter σ_i costs far commutations for $i \leq k-2$, nothing for $i = k-1$, one cancellation for $i = k$, and far commutations plus one braid move for $i \geq k+1$, and then recurse on h . A braid move $\sigma_i\sigma_{i-1}\sigma_i = \sigma_{i-1}\sigma_i\sigma_{i-1}$ costs four cells (three r_1 , one r_2). Charging each far commutation its actual width, the procedure uses 47 moves at total cost 323. The last claim follows by applying ψ to a product of conjugates of relators. $\square$

Proof of Theorem 3.1. Define words in $t^{\pm 1}, \rho^{\pm 1}$ by $W_1 = t$, $W_{2n} = W_n(t\rho t, R)$ and $W_{2n+1} = W_{2n}(t\rho, R')t$, and let $\text{Rw}(n) = \text{Area}(X_n W_n^{-1})$. Then $\text{Rw}(1) = 0$. Since $X_{2n} = \psi(X_n)$

literally, Lemma A.2 and the cheap-letter relations give $\text{Rw}(2n) \leq 864 \text{Rw}(n) + 84 |W_n|$ and $|W_{2n}| \leq 46|W_n|$. Since $X_{2n+1} = X_{2n}^b t$ freely, $\text{Rw}(2n+1) \leq \text{Rw}(2n) + 14 \cdot 46|W_n|$ and $|W_{2n+1}| \leq 7|W_{2n}| + 1$. Per binary digit, $\text{Rw} \leq 864 \text{Rw}' + 728|W'|$ and $|W| \leq 323|W'|$, so with $s = \lfloor \log n \rfloor$,

$$|W_n| \leq 323^s, \quad \text{Rw}(n) \leq 728 \sum_{i < s} 864^{s-1-i} 323^i \leq \frac{728}{541} 864^s \leq 1.346 n^{\log 864}.$$

Expanding $[\alpha, W_n]$ letter by letter, $U(n) \leq 2 \text{Rw}(n) + 5|W_n| \leq 2.70 n^{\log 864} + 5 n^{\log 323}$. For $m \geq 5$, $A(m) \leq 2U(m-2) + 18m - 35 \leq 5.39 m^{9.755} + 10 m^{8.336} + 18m \leq 0.87 m^{11}$, the ratio being largest at $m = 5$; and $A(2), A(3), A(4) \leq 1, 21, 47$. $\square$

Proof of Theorem 3.2. Put $q = \sigma_{-2}$ and add the auxiliary relation $r_6 = [q, b]$. It has a nine-cell filling in P , with counts $(3, 1, 0, 4, 1)$ in the original five types: two r_4 cells give $a^2 b a^{-2} = \sigma_1 \alpha b$, and commuting α through this word costs one adjacent braid (three r_1 cells and one r_2 cell) and one r_5 cell. Replacing the word on both sides costs four r_4 cells in total. We keep r_6 atomic until the final expansion. All recursive certificates use only the types $I = (1, 3, 4, 5, 6)$, so no image of r_2 is needed.

Here are finite certificates and their derivation, with vectors ordered by I . For $i \in I$, $\mathbf{e}_i$ denotes the unit vector for relator type r_i (thus $\mathbf{e}_3$ is the second coordinate vector, not the third). Let F_d fill $[\alpha, \sigma_d]$, C_m certify $\sigma_m = \tau_m$, and J_j certify $b\sigma_j b^{-1} = \sigma_{j+1}$. Starting from $\mathbf{f}_2 = \mathbf{e}_3$ and $\mathbf{c}_0 = 0$,

$$\mathbf{j}_j = 2j\mathbf{e}_4 + \mathbf{e}_5 + \sum_{d=2}^{j+1} \mathbf{f}_d, \quad \mathbf{c}_m = \mathbf{c}_{m-1} + \mathbf{j}_{m-1}, \quad \mathbf{f}_{m+2} = 2\mathbf{c}_m + 2m\mathbf{e}_6 + \mathbf{e}_3.$$

Indeed, $b\sigma_j b^{-1} = Q_j \sigma_{j+1} Q_j^{-1}$ with $Q_j = \sigma_0^{-1} \cdots \sigma_{j-1}^{-1}$ costs $2j$ copies of r_4 and one of r_5 ; commute past Q_j using distances $2, \dots, j+1$. For the last recurrence conjugate to $[q, \sigma_m]$, replace σ_m by τ_m , and move q past the $2m$ letters $b^{\pm 1}$ and then past α . This is noncircular. It gives

d	r_1	r_3	r_4	r_5	r_6
2	0	1	0	0	0
3	0	1	0	2	2
4	0	3	4	4	4
5	0	7	12	10	10
6	0	17	32	24	24

For $N_m = [\sigma_{-m}, b]$, $\mathbf{n}_2 = \mathbf{e}_6$ and $\mathbf{n}_m = \mathbf{n}_{m-1} + \mathbf{f}_{m-1} + 2\mathbf{e}_4$: conjugate by a^{-1} , replace $a^{-1}ba = \sigma_{-1}^{-1}b$, and remove the first factor using F_{m-1} . Thus $\mathbf{n}_3 = (0, 1, 2, 0, 1)$ and $\mathbf{n}_4 = (0, 2, 4, 2, 3)$.

Use the free substitution $D(a) = a^2, D(b) = b^2, D(\alpha) = T = [a^2, b^2]$. The equality $T = S := \sigma_1 \alpha \sigma_2 \sigma_1^{-1}$ has vector $\mathbf{v} = (1, 1, 4, 2, 0)$: expand into four commutators $[a, b]$, use r_5 twice, commute σ_2 past α , and invert the final σ_1 . The image counts are

$$\begin{aligned} \mathbf{d}_1 &= 2\mathbf{v} + \mathbf{e}_3 + 2\mathbf{e}_1, & \mathbf{d}_3 &= 4\mathbf{v} + \mathbf{f}_2 + 4\mathbf{f}_3 + 6\mathbf{f}_4 + 4\mathbf{f}_5 + \mathbf{f}_6, \\ \mathbf{d}_4 &= 0, & \mathbf{d}_5 &= 2\mathbf{v} + \mathbf{H}_0 + 2\mathbf{H}_1 + \mathbf{H}_2, \\ \mathbf{H}_j &= 2\mathbf{e}_4 + 2\mathbf{j}_j + \mathbf{f}_{j+2}, & \mathbf{d}_6 &= 2\mathbf{v} + 2(2\mathbf{n}_3 + \mathbf{n}_4 + \mathbf{n}_2). \end{aligned}$$

For $\mathbf{d}_1$, fill $(\alpha\sigma_2)^2$ inside its conjugator. For $\mathbf{d}_3$, the sixteen crossings between S and $a^4 S a^{-4}$ have gap multiplicities $1, 4, 6, 4, 1$. The fourth image is free. For the fifth, $h_2 := b^{-2}a^2 = t^2 q$ costs one r_4 cell, so $[h_2, \sigma_j]$ has vector $\mathbf{H}_j$, and $D(r_5) = b^2[h_2, T]b^{-2}$ freely. For the sixth,

replace $D(q)$ by $\sigma_{-3}\sigma_{-4}q\sigma_{-3}^{-1}$ and commute its four factors through b^2 . Consequently the substitution matrix is

$$M = \begin{pmatrix} 4 & 3 & 8 & 4 & 0 \\ 4 & 72 & 120 & 104 & 96 \\ 0 & 0 & 0 & 0 & 0 \\ 2 & 16 & 36 & 24 & 12 \\ 2 & 10 & 24 & 8 & 12 \end{pmatrix}.$$

Its rows are the counts of the chosen fillings of $D(r_i)$, not a substitution of the expanded nine-cell proof. For $w = (25, 616, 1, 133, 78)^T$, direct integer arithmetic gives

$$Mw = (2488, 65892, 0, 14070, 8234)^T \leq 107w.$$

Every weight dominates its original expansion cost, including $w_6 \geq 9$.

Put $u_i = t^{-i}qt^i$. The identities $\sigma_{-3} = u_1$, $\sigma_{-4} = u_2$ cost one and seven auxiliary cells: use r_6 for the first, then replace σ_{-3} twice in N_3 and conjugate the first identity by a^{-1} . On the formal t, q alphabet define

$$d(t) = t^2q, \quad d(q) = u_1u_2qu_1^{-1}, \quad E(t) = t^2qt^{-1}, \quad E(q) = q, \quad o = E \circ d.$$

These certify $D(t) = d(t)$, $D(q) = d(q)$, $b^{-1}D(t)b = o(t)$ and $b^{-1}D(q)b = o(q)$. Their respective cell vectors and reduced formal lengths are

rule	cell vector	length
$d(t)$	$(0, 0, 1, 0, 0)$	3
$d(q)$	$(1, 2, 6, 2, 6)$	10
$o(t)$	$(0, 0, 3, 0, 1)$	7
$o(q)$	$(1, 2, 12, 2, 10)$	26

The even rules follow from $h_2 = t^2q$ and the block identity; E expresses conjugation by b^{-1} , giving the odd rules. All four weighted costs are at most 2315.

Set $W_1 = t$, $W_{2m} = d(W_m)$ and $W_{2m+1} = o(W_m)t$, freely reducing in the formal alphabet. Substitution of cells, followed by these rules, certifies $h_k = b^{-k}a^k = W_k$: freely $h_{2m} = D(h_m)$ and $h_{2m+1} = b^{-1}D(h_m)bt$. If A_k is its weighted cost and $n_k = |W_k|$, then, with $m = \lfloor k/2 \rfloor$,

$$A_k \leq 107A_m + 2315n_m, \quad n_k \leq 26n_m + 1.$$

The t -exponent of W_m is m , so $n_m \geq 1$. Since $2315 + 29 \cdot 26 + 29 \leq 107 \cdot 29$,

$$A_k + 29n_k \leq 107(A_m + 29n_m) \leq 29 \cdot 107^{\lfloor \log k \rfloor} \leq 29k^{\log 107}.$$

Now $[t, \alpha]$ and $[q, \alpha]$ each cost one original cell. Replacing h_k twice and commuting its formal letters therefore fills $[h_k, \alpha]$ with at most $2A_k + n_k \leq 58k^{\log 107}$ original cells. The free identity $\sigma_m \tau_m^{-1} = b^m[h_m, \alpha]b^{-m}$ and the preceding far-commutator reduction give, for $k = m + 2 \geq 3$,

$$A(k) \leq 116m^{\log 107} + 18m + 1 \leq 135m^{\log 107}.$$

Together with $A(2) = 1$, this proves the theorem. Literal certificates for every image and rule above, the bounded-distance inputs and the recursive identities through $k = 8$ are distributed in `certificates107.json.gz`; an independent verifier recomputes their target words, their original-relator expansions and their cell histograms. The all-depth conclusion follows from the displayed recurrence, whose scalar induction is also checked in Lean. $\square$

Remark A.3 (Asymptotic refinement of this certificate scheme). Direct calculation gives $\det(xI - M) = x(x^4 - 112x^3 + 572x^2 + 3328x - 14208)$. The quartic changes sign on $(-6, -5), (3, 4), (7, 8), (106, 107)$, so the five eigenvalues are distinct and the spectral radius is $\rho \in (106, 107)$. Thus $\|M^j\| = O(\rho^j)$. Unweighted rule vectors are bounded by $(1, 2, 12, 2, 10)$, and $n_k \leq 27n_{\lfloor k/2 \rfloor}$. The resulting convolution is $O(\sum_{j < s} 27^j \rho^{s-1-j}) = O(\rho^s)$, since $\rho > 27$. Expanding auxiliary cells and commuting formal letters gives $A^*(M) = O(M^{\log \rho})$, where $\log \rho \approx 6.73252$. The explicit constants elsewhere use 107.

B From far commutators to the full Dehn function

Proof of Theorem 3.3. Fix an integer $n \geq 1$ and put $F_n = 1 + A^*(4n + 2)$ and $U(0) = 0$. Lemma A.1 gives

$$U(q) \leq \sum_{i=0}^{q-1} E(i) = O(n^2 F_n) \quad (0 \leq q \leq 2n).$$

The implied constants in this proof are absolute.

Buffered collection. Let w be a null word of length at most n and put $h = a^n b^n$. The two quotient coordinates while reading w after h stay in $[0, 2n]^2$. Write the current word as $v s(m, q)$, where $s(m, q) = a^m b^q$ and v is a word in the $\sigma_j^{\pm 1}$. Using $bab^{-1} = \alpha^{-1}a$ exactly q times gives $b^q ab^{-q} = \tau_{q-1}^{-1} \cdots \tau_0^{-1} a$. Replace each τ_j by σ_j at cost $U(j)$. Thus

$$\begin{aligned} s(m, q)a &= (\sigma_{m+q-1}^{-1} \cdots \sigma_m^{-1})s(m+1, q), \\ s(m, q)a^{-1} &= (\sigma_{m-1} \cdots \sigma_{m+q-2})s(m-1, q) \end{aligned}$$

each costs at most $q + \sum_{j < q} U(j) = O(n^3 F_n)$; an empty product is omitted. The second rule is the inverse rearrangement of the first at $(m-1, q)$, and is used only when the next coordinate is nonnegative. The rules $s(m, q)\alpha^{\pm 1} = \sigma_{m+q}^{\pm 1} s(m, q)$ cost $U(q)$, while multiplication by $b^{\pm 1}$ merely changes q and costs no cells. The buffer ensures that all indices produced are nonnegative.

After at most n rules the quotient coordinates return to (n, n) , and hence hwh^{-1} has been changed to a word v at cost $O(n^4 F_n)$. It has at most $2n^2 + n$ letters, with $0 \leq j \leq 4n$. Since $w = 1$, the word v is the identity permutation on the first $4n + 2$ points of ray 1. It remains to fill this permutation word with a cost that is uniform in the size of that symmetric group.

Coxeter insertion. In S_{d+1} , with adjacent transpositions $s_0, \dots, s_{d-1}$, use the recursive normal form uc_k , where $u \in S_d$, $c_k = s_{d-1}s_{d-2} \cdots s_k$ for $0 \leq k < d$, and $c_d = 1$. With right actions c_k sends the last point to $k+1$; this determines k , after which u is determined recursively. To insert a generator on the right use

$$c_k s_i = \begin{cases} s_i c_k, & i \leq k-2, \\ c_{k-1}, & i = k-1, \\ c_{k+1}, & i = k, \\ s_{i-1} c_k, & i \geq k+1. \end{cases}$$

The first case uses far commutations, the middle cases concatenate or cancel, and the last case moves the final s_i left across $s_k, \dots, s_{i-2}$, applies one braid, and moves the resulting s_{i-1} left across $s_{i+1}, \dots, s_{d-1}$. There are at most $d+1$ Coxeter moves at a level and at most d levels per insertion, since only one generator is passed into u . Consequently a word of length l reduces to its unique normal form in $O(ld^2)$ moves, and the identity reduces to the empty word.

Apply this with $d = 4n + 1$ to v . First replace inverse letters by positive ones using $\sigma_j^2 = 1$, at one cell per letter. A Coxeter square costs one cell in P , a braid costs at most four, and a far commutation costs at most $A^*(4n + 2)$. Thus the $O(n^2)$ letters of v fill at cost $O(n^4 F_n)$. Combining with collection and using invariance of area under free conjugation proves the asserted bound for $\delta_P(n)$. Substituting Theorem 3.2, or Theorem 3.1, gives the two polynomial exponents. $\square$

Use of AI assistance

Generative AI tools assisted with developing arguments and constructions, checking proofs and numerical identities, searching the literature, writing verification code, and drafting and editing the manuscript. Automated review is not a substitute for mathematical proof. The accompanying verification notes distinguish executable checks and kernel-checked Lean results from proofs in the manuscript that are not formalized in Lean. The authors take responsibility for the content.

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